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SUPERSEDING
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MILITARY SPECIFICATION

MULTIPLEXER, TD-1233(P)/TTC

This specification is approved for use by the Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the design, fabrication and testing necessary for the manufacture of the Multiplexer,TD-1233(P)/TTC. This unit is also known as the Remote Loop Group Multiplexer (RLGM) and is one of the electronic units that comprise the Digital Group Multiplex (DGM) family of cable system equipment developed by the Joint Tactical Command, Control and Communications Agency (JTC^3A, formerly TRI-TAC). See 6.1 for intended use and functional description of the unit.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, US Army Communications-Electronics Command, ATTN: AMSEL-ED-LC, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC No. N/A FSC 5820

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SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-P-11268	Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-C-55583	Cable Assembly, Special Purpose CX-11230( )/G, Cable Assembly Adapter CX-10734( )/G
MIL-E-55585	Electronics Equipment and Parts, Packaging of

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired and Other Devices
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461A Notice 4	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462 Notice 3	Electromagnetic Interference Characteristics Measurement of
MIL-STD-781C	Reliability Tests, Exponential Distribution
MIL-STD-810D	Environmental Test Methods and Engineering Guidelines
MIL-STD-1332	Definitions of Tactical, Prime, Precise, and Utility Terminologies for Classification of the DOD Mobile Electric Power Engine Generator Set Family
MIL-STD-2073-2	Packaging Requirements Code
MIL-STD-45662	Calibration System Requirements

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

SPECIFICATION

Communications-Electronics Command

CR-CX-0173-001	Nuclear Survivability/Vulnerability Provisions for the Family of Digital Group Multiplexer Equipments

DRAWINGS

Communications-Electronics Command

DL-SM-B-876631	Loop Group Multiplexer, Remote TD-1233(P)/TTC
SM-D-884567	Circuit Card Assembly (CCA), Multi Vdc
SM-D-884571	CCA, 52 Vdc
SM-D-884575	CCA, DC Input
SM-D-884579	CCA, AC Input
SM-D-959323	CCA, Analog Applique Unit (AAU)
SM-E-986232	CCA, Group Modem
SM-E-986235	CCA, Loop Modem
SM-E-986239	CCA, Timing Generator
SM-E-986241	CCA, BITE 2
SM-E-986247	CCA, MUX/DEMUX
SM-E-986250	CCA, Group Framing

(Copes of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article units. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction.

3.2.1 Fabrication. The equipment shall be constructed in accordance with DL-SM-D-876631 and the requirements specified herein.

3.2.2 Printed wiring. The printed wiring shall meet the requirements of MIL-STD-454, Requirement 17.

3.2.3 Semiconductor devices and integrated circuits. The semiconductor devices and integrated circuits shall meet the requirements of MIL-STD-454, Requirements 30 and 64.

3.3 Parts, materials, and processes-general. In addition to the requirements of this specification, the requirements of MIL-P-11268 including the selection requirements therein shall apply.

3.4 Finish. The equipment shall be finished in accordance with MIL-F-14072 and the equipment drawing DL-SM-D-876631.

3.5 Marking. The marking shall conform to MIL-M-13231, with approvals in accordance with requirements specified in the contract or purchase order.

3.5.1 Serial numbers. (See 4.15) A unique serial number is required on each nomenclatured item and CCA. The numbers shall be consecutively assigned.

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3.6 Electrical preconditioning (burn-in). (See 4.6) Prior to submission to the specified quality conformance inspections, each assembled unit shall be electrically preconditioned for a minimum of 48 hours operating time. The last 24 hours of electrical preconditionings shall be failure free. Each unit fabricated shall be subjected to this process. Failure free is defined as (1) no built-in test equipment (BITE) alarms during burn-in and (2) the successful completion of the operational test of 4.7.12 after burn-in.

3.7 Electrical. (See 4.7)

3.7.1 Diphase group output. (See 4.7.1) The conditioned diphase group output signal shall transmit data at 72 or 144 kb/s and have the following characteristics.

3.7.1.1 Output voltage. (See 4.7.1.1) The diphase group output voltage shall be 3 volts ±20% peak-to-peak into a 58-ohm resistive load.

3.7.1.2 Output rise and fall time. (See 4.7.1.2) The diphase group output rise and fall time shall be 130 ns maximum (10 to 90%) into a 57.6-ohm ±1% resistive load.

3.7.1.3 Output impedance. (See 4.7.1.3) The diphase group output impedance shall be 58 ohms ±15% from 1000 Hz to 300 kHz, and 50 to 100 ohms from 300 to 1000 Hz.

3.7.2 Diphase group input. (See 4.7.2) The diphase group input shall accept diphase signals at 144 kb/s or 72 kb/s and have the following characteristics.

3.7.2.1 Input impedance. (See 4.7.2.1) The diphase group input impedance shall be 58 ohms ±15% from 1000 Hz to 300 kHz, and 50 to 100 ohms from 300 to 1000 Hz.

3.7.2.2 Received signal level. (See 4.7.2.2) The unit shall operate properly on the signal specified in 3.7.1 at the end of 2 miles of CX-11230 (see MIL-C-55583) cable with an additional 4 dB of attenuation. Proper operation shall be defined as a channel bit error rate (BER) of zero errors in 30 seconds.

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3.7.3 Station clock input. (See 4.7.3) The input shall accept a balanced input at the Group bit rate.

3.7.3.1 Receiver dynamic range. (See 4.7.3.1) The receiver shall deliver the proper digital output when a line-to-line voltage of +50 mV to 4 V for a digital 1 or -50 mV to -4 V for a digital 0 is applied to the receiver input terminals.

3.7.3.2 Input load impedance. (See 4.7.3.2)The input load impedance shall be 100 ohms ±10% line-to-line. The resistive termination shall float: i.e., the center tap is not grounded.

3.7.3.3 Timing modes.(See 4.7.3.3)

3.7.3.3.1. Loop back mode. Transmit timing shall be derived from the received group signal. The multiplexer shall accept external timing frequency errors up to ±50 PPM from the nominal.

3.7.3.3.2. Station clock mode. Transmit timing shall be derived from an external timing source. The received group signal shall be slaved to this timing source by external means. When the external timing signal is absent, the transmit timing shall be derived from an internal timing source. The frequency of this timing source shall be within ±40 ppm of the nominal frequency.

3.7.4 Orderwire interface. (See 4.7.4) The equipment shall have only a passive analog orderwire interface with the following characteristics.

3.7.4.1 Input impedance.(See 4.7.4.1) The orderwire interface input impedance shall be 600 ohms ±10% at a frequency of 1 kHz.

3.7.4.2 Output impedance. (See 4.7.4.2) The orderwire interface output impedance shall be 600 ohms ±10% at a frequency of 1 kHz.

3.7.4.3 Receive level.(See 4.7.4.3) The orderwire receive level shall be between -13 and -10 dBm with a +21.3 dBm input level at 1.6 kHz. The diphase group output shall be looped through less than 5 feet of cable.

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3.7.4.4 Bandwidth. (See 4.7.4.4) The frequency response of an orderwire link, as described in 3.7.4.3 shall be as follows:

Frequency	Transmission Loss (from reference)
300	-5 to +8 dB
400	±5 dB
1000	Reference
3000	±5 dB
3400	-5 to +8 dB

3.7.4.5 Signal-to-noise-ratio. (See 4.7.4.5) With random group traffic the signal-to-noise ratio C-MESSAGE shall be greater than 11 dB at 72 kb/s and 20 dB at 144 kb/s.

3.7.5 Digital channel characteristics. (See 4.7.5)

3.7.5.1 Output voltage. (See 4.7.5.1) The output voltage shall be 3 volts ± 15% peak-to-peak measured into a 125 ohm load.

3.7.5.2 Output waveshape. (See 4.7.5.2) Measured into a 125 ohm resistive load, the rise and fall times(10% to 90%) shall be between 0.65 and 3.0 microseconds. The droop shall be less than 5% of the peak-to-peak amplitude.

3.7.5.3 Input and output impedance. (See 4.7.5.3) The input and output impedance magnitudes shall be 125 ohms ±15% over the band from 8 to 56 kHz.

3.7.5.4 Input voltage. (See 4.7.5.4) Proper operation shall occur over the input range of 0.15 to 4.0 volts peak-to-peak. Proper operation shall be defined as a channel bit error rate (BER) of zero errors in 30 seconds.

3.7.5.5 Idle loop. (See 4.7.5.5) When transmission from the DSVT has ceased on a loop, the loop modem of the equipment shall send continuous ones on its equipment side.

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3.7.6 Loop power feed. (See 4.7.6 and 4.7.13.4) Maximum output voltage shall be 56 Vdc into a 11,200-ohm resistive load. Minimum output current shall be 34.9 mA into a 1373 ohm resistive load. The loop outputs shall be short circuit-proof, both line-to-line and line-to-ground. The power feed circuitry shall tolerate an initial current surge of 100 ma for up to 45 milliseconds without power feed shut off.

3.7.7 Analog channel characteristics. (See 4.7.7)

3.7.7.1 Analog channel input/output. (See 4.7.7.1) The impedance at the 4-wire analog channel input and output terminals shall be 600 ohms resistive with a minimum return loss of 26 dB over the frequency range 300 to 3400 Hz.

3.7.7.2 Longitudinal balance. (See 4.7.7.2) The longitudinal balance at the analog channel input and output terminals shall be not less than 40 dB when measured at a tone level of 0 dBm over the frequency range of 300 to 3400 Hz.

3.7.7.3 Frequency response. (See 4.7.7.3) The characteristics of the input/output filter should be as follows:

*Frequency, Hz	Input Atten (dB)	Output Atten (dB)
76	6 min	6 min
200	3 ± 1	3 ± 1
1000	ref	ref
3500	3 ± 1	0 to 4
8800	12 min	45 min

3.7.7.4 Analog channel input and output test tone level and frequency. (See 4.7.7.4) The nominal test tone input and output level shall be -17 dBm at 820 Hz.

3.7.7.5 Input-output linearity. (See 4.7.7.5) The output level of each channel (See 3.7.7.10.3 and 3.7.7.10.4) shall be within ±1.0 dB of the input signal level from full load tone (see 3.7.7.9) to 35 dB below full load tone when the input and output level selector switches are set to a sum of 0 dB.

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3.7.7.6 Loopback switch. (See 4.7.7.6) A loopback switch shall be provided for each channel. It shall permit the digital output of the encoder to be looped to the digital input of the decoder. The loopback switches shall be operable when the modules are inserted into the equipment.

3.7.7.7 Noise. (See 4.7.7.7)

3.7.7.7.1 Idle channel noise. (See 4.7.7.7.1) With one or more analog modules inserted in any unit and the remaining module positions filled with digital modules and the analog channels operating in the loopback mode, the signal measured on the analog inputs and outputs shall not exceed 33 dBrnC when all analog ports are terminated in 600 ohms and input and output level selector switches are set to 0 dB.

3.7.7.7.2 Crosstalk. (See 4.7.7.7.2) The inter-channel and intra-channel crosstalk plus noise measured at either the near end or the far end of two AAU channels shall not exceed 34 dBrnC when sending the 820 Hz test tone at full load.

3.7.7.8 Peak clipping. (See 4.7.7.8) For a +4 dBm input signal at 200 Hz, the output signal shall have a total distortion of less than 10% at a channel rate of 32 kb/s and with the input level selector switch set to -10 dB and the output level selector switch set to +10 dB.

3.7.7.9 Full load tone. (See 4.7.7.9) The nominal full load tone input and output level shall be 0 dB at a frequency of 820 Hz with the input level selector switch set to -10 dB, and the output level selector switch set to +10 dB. With a full load tone, the Continuous Variable Slope Delta (CVSD) converter approaches the slope overload condition.

3.7.7.10 Analog channel signal processing. (See 4.7.7.10)

3.7.7.10.1 Clock rate. (See 4.7.7.10.1) The channel clocking rate of the CVSD converter shall be 16 or 32 kb/s, selectable.

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3.7.7.10.2 Dynamic range. (See 4.7.7.10.2) The dynamic range of the CVSD shall be such that the repetitive test patterns shown below shall produce 666 2/3 Hz tone outputs having a ratio of 23 ± 1 dB.

Channel Rate kb/s	Digital Pattern (Hexadecimal Equivalent)*											
32 low output	D	D	5	2	4	9	2	2	A	D	B	6
high output	A	E	F	F	F	B	5	1	0	0	0	4
16 low output	6	E	A	9	2	5						
high output	B	F	D	4	0	2						

3.7.7.10.3 Output level control. (See 4.7.7.10.3) The output level of the CVSD decoder shall be selectable in four steps independent of the level setting at the encoder. The output level selector shall be labeled -3, 0, +10, and +20. With either of the two repetitive digital patterns (shown below) applied to the decoder, the analog output into a 600 ohm load shall be -20, -17, -7, or +3 dBm ±0.5 dB, when the output level selector is set at -3, 0, +10, or +20 respectively.

Channel Rate kb/s	Digital Pattern (Hexadecimal Equivalent)*									
32	F	D	A	A	1	0	2	5	5	E
16	F	B	4	1	2					

*The serial bit patterns are read from left to right and are expressed as hexadecimal characters; each character is transmitted in time sequence as read, MSB first and LSB last.

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3.7.7.10.4 Input level control. (See 4.7.7.10.4) The operating level of the CVSD encoder shall be adjustable in four steps. The input level selector shall be labeled +3, 0, -10, and -20. When the channel is in the loopback mode, the input levels and level settings shall produce the output levels shown below with a tolerance of ± 1.0 dB.

Input Level (dBm)	Encoder Level SETTING	Decoder Level SETTING	Output Level (dBm)
-20	+3	+20	+3
-17	0	+20	+3
- 7	-10	+20	+3
+ 3	-20	+20	+3
-20	+3	+10	-7
-17	0	+10	-7
- 7	-10	+10	-7
+ 3	-20	+10	-7
-20	+3	0	-17
-17	0	0	-17
- 7	-10	0	-17
+3	-20	0	-17
-20	+3	-3	-20
-17	0	-3	-20
- 7	-10	-3	-20
+ 3	-20	-3	-20

3.7.7.11 Envelope delay distortion. (See 4.7.7.11) The envelope delay relative to 800 Hz of the input and output filters added together shall be less than 100 us from 800 to 3000 Hz.

3.7.8 Framing. (See 4.7.8)

3.7.8.1 Format. (See 4.7.8.1) The overhead channel shall be channel zero of the equipment group as shown in figure 1. The overhead channel shall be composed of four equal-rate subchannels of 4 kb/s when the channel bit rate is 32 kb/s and 2 kb/s when the channel bit rate is 16 kb/s. Subchannel zero shall be the frame subchannel and subchannel one shall be the telemetry subchannel. Subchannels two and three shall be spare subchannels and shall be set to one.

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3.7.8.2 Frame acquisition time. (See 4.7.8.2) The frame acquisition time is defined as the period from the start of frame search until the multiplexer at each end of a link is capable of restoring traffic in both directions on the link between them. The equipment shall acquire frame synchronization within 50/100 ms at frame rates of 4/2 kb/s, with a minimum probability of 0.9, in an error environment of 0.1% BER random errors. The equipment shall also acquire frame synchronization within 100/150 ms at frame rates of 4/2 kb/s, with a minimum probability of 0.9, in an error environment of 20% BER random bursts interspersed with 0.1% BER random error intervals. The time between the start of successive bursts shall be random between 50 ms and 1 second, and the burst length shall be 5% of the time between the start of that burst and the start of the next burst. The above times do not include cryptographic resync delays, or any preset delay intended to assist in flywheeling through fades.

3.7.8.3 Frame maintenance. (See 4.7.8.3) While bit count integrity is maintained "out of sync" shall not be declared more often than once per 240 hours (on the average) in an error environment of 20% BER burst interspersed with 0.1% BER random error intervals. The time between the start of successive bursts shall be random between 50 ms and 1 second, and the burst length shall be 5% of the time between the start of that burst and the start of the next burst.

3.7.8.4 Detection of out of sync. (See 4.7.8.4) When bit count integrity is lost and with random data on all channels, the equipment shall detect out of sync within 150 ms at frame rates of 4/2 kb/s, with a minimum probability of 0.9.

3.7.8.5 Idle filler pattern. (See 4.7.8.5) When the equipment is out of frame it shall set its channel outputs to all ones.

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3.7.9 Prime power. (See 4.7.9) The dc primary power shall have a steady-state voltage of 28 (+4, -6) volts. The ac primary power source shall be 115 volts ±10%, single phase. The frequency of the ac voltage shall be 50 Hz -5% to 400 Hz +5%. The ac primary power shall conform to MIL-STD-1332 for Type I, Class 1 and 2, Mode I, II, and III generator set. The input power to the equipment shall not exceed 60 watts, except that it shall not exceed 42 watts when operating on cable power feed as follows:

The equipment shall operate on power feed over the CX-11230 cable having the following characteristics: Positive polarity shall appear on both center conductors, negative on both shields; voltage shall range from 130 to 196 Vdc at the input of the power supply. Input power limits shall be measured with all four loop power feeds loaded with 1373 ohms.

3.7.10 Current protection. (See 4.7.10) The cable interface circuits shall withstand without damage to the equipment an injected current of 90 mA on either center conductor of the CX-11230 cable.

3.7.11 BITE and alarms. (See 4.7.11) The BITE shall monitor signals in and out of the unit and internal circuits. Go/no-go fault location circuits shall isolate failures to a particular printed circuit card or lowest replaceable unit (LRU). Indicating devices shall be provided to ensure proper operation of the equipment and to monitor the dc voltages, go/no-go fault-isolation, and other critical internal and external signals. The BITE shall operate without interrupting traffic or interfering with the normal system operation. Removal of the BITE card shall not affect operation of the unit. Failure alarms shall be provided for channel failure, common power supply failure, and inter-unit signal malfunctions.

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3.7.11.1 BITE alarms. (See 4.7.11.1) Failures detected by the BITE circuity cause relay contact closures accessible externally at a connector on the rear panel. The S, SA and LOI alarm outputs described below shall be available.

a. S alarm. The S alarm shall indicate any failure detected including loss of input power. This alarm shall also provide a visible indication on the front panel except for loss of power.

b. SA alarm. The SA alarm shall indicate that the malfunction is within the unit. If the unit power switch is set to OFF, there shall be no SA alarm.

c. LOI alarm. The LOI alarm shall indicate a loss of input on the group side of the unit.

3.7.12 Operational check. (See 4.7.12)

3.7.12.1 Digital channel check. (See 4.7.12.1) There shall be no errors for a period of 30 seconds with the orderwire on.

3.7.12.2 Maintenance orderwire check. (See 4.7.12.2) The orderwire output level shall be between -13 and -10 dBm with a +21.3 dBm input level at 1.6 kHz.

3.7.12.3 DC voltage check. (See 4.7.12.3) The loop channel outputs, when enabled, shall provide power to the other DSVT at a voltage between 46.9 Vdc and 56 Vdc.

3.7.12.4 Sync/frame check. (See 4.7.12.4) The unit shall reframe when the power supply is interrupted and turned back on.

3.7.12.5 Analog channel check. (See 4.7.12.5) With all level switches set to 0, the gain shall be 0 dBm $\pm$1 dB at 820 Hz.

3.7.12.6 Alarm monitor check. (See 4.7.12.6) There shall be no alarms generated when the unit is operated normally. The S and LOI (but not SA) alarm shall be actuated when the group cable is removed from the unit under test(UUT). The alarm state is defined as a closed set of relay contacts in each of the S and SA connections and illumination of the LOI indicator.

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3.7.13 Environmental operational check. (See 4.7.13)

3.7.13.1 Digital channel check. (See 4.7.13.1) There shall be no errors for a period of 30 seconds with the orderwire on.

3.7.13.2 Maintenance orderwire check. (See 47.13.2) The orderwire output level shall be between -13 and -10 dBm with a +21.3 dBm input level at 1.6 kHz.

3.7.13.3 DC voltage check. (See 4.7.13.4) The loop power feed voltage at each of the digital channels shall be not less than 46.9 Vdc.

3.7.13.4 Sync/frame check. (See 4.7.13.4) The unit shall reframe when the power supply is interrupted and turned back on.

3.7.13.5 Analog channel check. (See 4.7.13.5) With all level switches set to 0, the gain shall be within ±1 dB of the room temperature value at 820 Hz

3.7.13.6 Alarm monitor check. (See 4.7.13.6) When the UUT is operated normally there shall be no alarms generated. Depressing the test button on the UUT shall cause S, SA and LOI alarms. The alarm state is defined as a closed set of relay contacts in each of the S and SA connectors and illumination of the LOI indicator.

3.8 Service conditions. (See 4.8) The operational test of 3.7.13 shall be performed before, during, and after each service condition test unless otherwise specified in the appropriate test paragraph.

3.8.1 Temperature. (See 4.8.1)

3.8.1.1 High temperature. (See 4.8.1.1) The equipment shall be operable without degradation in specified performance at ambient air temperatures as high as +51.7°C (+125°F) plus the heating effects of solar radiation at a rate of 360 BTU/ft^2/hr. The equipment shall withstand exposure (non-operating) to ambient air temperatures as high as +71.1°C (+160°F).

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3.8.1.2 Low temperature. (See 4.8.1.2) The equipment shall be operable without degradation in specified performance at ambient temperatures as low as -45°C (-50°F). The equipment shall withstand exposure (non-operating) to ambient air temperatures as low as -57°C (-70°F).

3.8.2 Humidity. (See 4.8.2) The equipment shall be operable without degradation in specified performance, and shall sustain no physical damage during and after prolonged exposure to extreme high humidity levels as encountered in tropical areas.

3.8.3 Low pressure (altitude). (See 4.8.3) The equipment shall be operable without degradation in specified performance at altitudes up to 10,000 feet above sea level, and shall withstand air transportation at altitudes up to 40,000 feet above sea level.

3.8.4 Dust. (See 4.8.4) The equipment shall withstand, in both operating and non-operating conditions, exposure to fine dust particles with wind speeds of 3542 ft/min (35 knots).

3.8.5 Salt fog. (See 4.8.5) The equipment shall be resistant to the corrosive effects of salt fog atmosphere.

3.8.6 Fungus. (See 4.8.6) The equipment, in the assembled-and-ready-for-delivery condition, shall provide no nutrients in material, coating or contaminant form or support fungal growth when tested in accordance with 4.8.6. Only inherently fungus resistant grades of materials per requirement 4, MIL-STD-454, shall be used.

3.8.7 Immersion. (See 4.8.7) Prior to and following vibration and shock testing, the equipment in its transit case shall be immersion-proof to a covering depth of 3 feet of water, for a period of not less than 2 hours.

3.8.8 Vibration and shock. (See 4.8.8) The equipment shall withstand vibration and shock induced during field transport by military vehicle and by rough handling. The equipment shall withstand vibrations encountered during transportation and shocks encountered during servicing.

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3.8.9 Rain. (See 4.8.9) The equipment shall be operable without degradation in specified performance, and shall sustain no physical damage during periods of heavy precipitation with intermittent wind.

3.9 Electromagnetic interference (EMI). (See 4.9) The TD-1233()/TTC shall meet the following emission and susceptibility requirements of MIL-STD-461A, Notice 4:

CE01	CS01	RE02	RS02
CE02	CS02	RE02.1	RS03
CE03	CS06		RS03.1
CE04			
CE05			

NOTES:

a. RE02.1 shall be applicable over the frequency range of 14 kHz to 1000 MHz.

b. RS03 shall be applicable over the frequency range of 10 kHz to 1000 MHz.

c. Conducted emission and susceptibility requirements are applicable to dc, 60 Hz and 400 Hz power lines.

3.9.1 Bonds and grounds. Bonding and grounding impedance shall not exceed 2.5 milliohms.

3.10 Reliability. (See 4.10) The equipment shall demonstrate a specified mean-time-between-failures (MTBF), θ_o of 4000 hours when operating in the normal mode.

3.11 Interchangeability. (See 4.11) Like units, assemblies, subassemblies, and replaceable parts shall meet Requirement 7 of MIL-STD-454.

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3.12 Unit subassembly electrical performance. (See 4.12) The unit subassemblies listed below shall meet the requirements specified on their respective test requirement specification (TRS) drawings, as referenced in the following subassembly drawings:

CCA's	Drawing
Multi Vdc	SM-D-884567
52 Vdc	SM-D-884571
DC Input	SM-D-884575
AC Input	SM-D-884579
Group Modem	SM-E-986232
Loop Modem	SM-E-986235
Timing Generator	*SM-E-986239
BITE 2	*SM-E-986241
Mux/Demux	*SM-E-986247
Group Framing	*SM-E-986250
Analog Applique Unit (AAU)	SM-D-959323

*This drawing contains automatic test equipment (ATE) acceptance procedures and programs.

3.13 Operational test. (See 4.7.12) Each fabricated and tested equipment shall be subjected to a final operational test in accordance with 4.7.12 after all other inspections and tests are completed to determine that the equipment is operational prior to packaging.

3.14 Systems safety engineering. (See 4.13)

3.14.1 Personnel hazards. Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, Requirements 1 and 8 shall be made applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly processes, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with this requirement has been achieved shall be through a visual examination and inspection.

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3.14.2 Rounding exposed edges. Exposed edges and corners shall be rounded sufficiently to preclude puncture or laceration hazards.

3.14.3 Radioactive materials. Radioactive matrials shall not be used (e.g., luminous dials/markings, electron tubes, surge arrestors and lenses).

3.15 Workmanship. (See 4.13) The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and with the applicable portions of MIL-P-11268 as follows:

a. Adhesives

b. Cables, cords and wires (electrical)

c. Wiring (harnessing)

d. Plastic material and parts

e. Rubber

f. Wiring and cabling

g. Connectors (electrical)

h. Printed wiring assemblies

i. Processes

3.16 Nuclear survivability. (See 4.16) The equipment shall satisfy the nuclear survivability requirements stated in Section 3 of CR-CX-0173-001.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection. (See 4.3) Does not include packaging.

b. Inspection covered by subsidiary documents. (See 4.4)

c. Quality conformance inspections.

(1) Quality conformance inspection of equipment before packaging. (See 4.5)

(2) Packaging inspection. (See 4.14)

4.3 First article inspection. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish a minimum of 20 first article units of the complete equipment. The first article quantity shall be as specified in the contract or purchase order.

4.3.2 First article inspection. The first article inspection shall consist of inspections specified in table I and shall be performed in the order specified therein except that all hardwired assemblies shall undergo inprocess group A inspection prior to burn-in.

4.3.3 First article data. The first article test plan and test report(s) shall be as required in the contract or purchase order.

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TABLE I. First article inspection.

Inspection	Rqmt Para	Test Para	Unit 1	2	3	4	5	6	7	8	9	10	11	12-19	20+
1. Inspections by Subsidiary Documents	N/A	4.4	Inspection to be performed on all units.												
2. Burn-In	3.6	4.6	Inspections to be performed on all units.												
3. Group A Inspection	(See Table II)		Inspections to be performed on all units.												
4. Group B Inspection	(See Table III)		Inspections to be performed on all units.												
5. Group C Inspection															
Subgroup 1															
High temperature	3.8.1.1	4.8.1.1	1	*											
Low temperature	3.8.1.2	4.8.1.2	2	*											
Subgroup 2															
EMI	3.9	4.9			1										
Immersion	3.8.7	4.8.7				1	*								
Vibration/shock	3.8.8	4.8.8				2									
Immersion	3.8.7	4.8.7				3									
Subgroup 3															
Humidity (Note c)	3.8.2	4.8.2						1			*				
Rain	3.8.9	4.8.9						2			*				
Subgroup 4															
Altitude	3.8.3	4.8.3							1		*				
Dust (Note c)	3.8.4	4.8.4							2		*				
Salt fog (Note c)	3.8.5	4.8.5							3						
Fungus (Note c)	3.8.6	4.8.6								1					
Subgroup 5															
Output rise/fall times	3.7.1.2	4.7.1.2										1			
Output impedance	3.7.1.3	4.7.1.3										2			
Input imedance	3.7.2.1	4.7.2.1										3			

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TABLE I. First article inspection (Continued)

Inspection	Rqmt Para	Test Para		Unit 1	2	3	4	5	6	7	8	9	10	11	12-19	20+
Receiver dynamic range	3.7.3.1	4.7.3.1											4			
Input load impedance	3.7.3.2	4.7.3.2											5			
Timing modes	3.7.3.3	4.7.3.3											6			
Orderwire Input impedance	3.7.4.1	4.7.4.1											7			
Orderwire Output impedance	3.7.4.2	4.7.4.2											8			
Orderwire Receive level	3.7.4.3	4.7.4.3											9			
Bandwidth **	3.7.4.4	4.7.4.4											10			
Signal to noise ratio **	3.7.4.5	4.7.4.5	(D)										11			
Output wave shape	3.7.5.2	4.7.5.2	(D)										12			
Input and output impedance	3.7.5.3	4.7.5.3	(D)										13			
Idle loop **	3.7.5.5	4.7.5.5	(D)										14			
Analog channel input/output Impedance	3.7.7.1	4.7.7.1	(A)										15			
Longitudinal balance	3.7.7.2	4.7.7.2	(A)										16			
Peak clipping **	3.7.7.8	4.7.7.8	(A)										17			
Format **	3.7.8.1	4.7.8.1	(D)										18			
Frame aquisition time	3.7.8.2	4.7.8.2											19			
Frame maintainance **	3.7.8.3	4.7.8.3											20			
Detection of out-of-sync	3.7.8.4	4.7.8.4											21			
Idle filler pattern **	3.7.8.5	4.7.8.5	(D)										22			
Prime power	3.7.9	4.7.9	(D)										23			
Current protection **	3.7.10	4.7.10	(D)										24			
Interchange ability	3.11	4.11											25			
Crosstalk	3.7.7.7.2	4.7.7.7.2											26			
Loop power feed	3.7.6	4.7.6	(D)										27			
Loopback switch **	3.7.7.6	4.7.7.6											28			
Envelope Delay Distortion **	3.7.7.11	4.7.7.11											29			
6. Group D Inspection																
Reliability	3.10	4.10													8 Units	

(A) Analog Applique CCA in card slots A6 and A7.
(D) Loop Modem CCA in card slots A6 and A7.
** One time engineering test covered by a separate engineering report. Not a part of the first article report.

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NOTES for table I.

a. The table I inspections 1 through 4, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements specified in the table.

b. The numbers in the unit columns in the table covered by inspections 5 and 6 specify the order of inspections for the indicated unit. An asterisk (*) indicates the use of an auxiliary unit (a TD-1233 production unit).

c. The equipment shall be thoroughly washed, cleaned, dried, and refurbished after this inspection before proceeding with subsequent inspection.

d. All first article units shall be subjected to group A and group B inspections.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

	Where required
Printed wiring	3.2.2
Semiconductor devices and integrated circuits	3.2.3
Parts, material, and processes	3.3
Finish	3.4
Marking	3.5

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.4,and 4.5.1 through 4.5.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

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4.5.1 Group A inspection. Each unit on contract or purchase order and each subassembly and assembly identified in table II shall be inspected for conformance to the inspections specified in table II.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in the order listed in table II. The order of inspection of the subassemblies, assemblies, and units (and equipment) shall be as follows:

a. Prior to their assembly into the end equipment, the assemblies and subassemblies identified in table II shall be subjected to a 100% visual/mechanical and electrical inspection by the contractor for conformance with the inspection requirements provided in table II.

b. Only for spares which are not installed into complete units, do the following: After completion of the above 100% inspection by the contractor, discrete lots shall be formed from the subassemblies and assemblies which passed the 100% inspection and these shall be subjected to sampling inspection utilizing the procedures of MIL-STD-105 using general inspection levels and AQLs indicated in table II (Quality Assurance Verification). Factors of lot composition not defined herein or in the contract or purchase order, shall be in accordance with MIL-STD-105.

c. Assemblies and subassemblies which have passed the 100% visual/mechanical and electrical inspection of step a above shall be assembled into the end equipments, which shall then be subjected by the contractor to the burn-in test requirements of 4.6 and an end equipment 100% visual/mechanical and electrical inspection in accordance with table II, in that order.

d. After completion of the above 100% inspection by the contractor, discrete lots shall be formed from units which passed the 100% inspection and these shall be subjected to sampling inspection by the contractor utilizing the procedures of MIL-STD-105 using the gerneral inspection levels and AQLs indicated in table II. Group A electrical inspection shall be performed in the order shown in table II.

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TABLE II. Group A inspection.

Inspection	Reqt Para	Insp Para	AQL Major	AQL Minor
VISUAL-MECHANICAL (Including spares)				
TD-1233 Case and Panels	3.15	4.13	2.5%	10.0%
Multi Vdc	3.15	4.13	2.5%	4.0%
Timing Generator	3.15	4.13	2.5%	4.0%
BITE 2	3.15	4.13	2.5%	4.0%
Loop Modem	3.15	4.13	2.5%	4.0%
MUX/DEMUX	3.15	4.13	2.5%	4.0%
Group Framing	3.15	4.13	2.5%	4.0%
Group Modem	3.15	4.13	2.5%	4.0%
Analog Applique Unit (AAU)	3.15	4.13	2.5%	4.0%
52 Vdc	3.15	4.13	2.5%	4.0%
DC Input	3.15	4.13	2.5%	4.0%
AC Input	3.15	4.13	2.5%	4.0%
ELECTRICAL PERFORMANCE				
Timing Generator	3.12	4.12	2.5%	*
BITE 2	3.12	4.12	2.5%	*
Loop Modem	3.12	4.12	2.5%	*
MUX/DEMUX	3.12	4.12	2.5%	*
Group Framing	3.12	4.12	2.5%	*
Group Modem	3.12	4.12	2.5%	*
Analog Applique Unit (AAU)	3.12	4.12	2.5%	*
52 Vdc	3.12	4.12	2.5%	*
DC Input	3.12	4.12	2.5%	*
AC Input	3.12	4.12	2.5%	*
Multi Vdc	3.12	4.12	2.5%	*
ELECTRICAL				
Operational tests	3.7.12	4.7.12(M)	1.0%	*
Idle channel noise	3.7.7.7.1	4.7.7.7.1(M)	1.0%	*
BITE and Alarms	3.7.11	4.7.11 (M)	1.0%	*

(M) Analog Applique Unit (AAU) CCA in card slot A6 and Loop Modem CCA in card slot A7.

*All electrical defects are major.

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4.5.2 Group B inspection. Group B inspection shall be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. This inspection shall conform to table III and to the special levels of table I of MIL-STD-105.

TABLE III. Group B. inspection.

Inspection	Reqt Para	Insp Para	AQL
Output voltage	3.7.1.1	4.7.1.1	4.0%
Received signal level	3.7.2.2	4.7.2.2	4.0%
Output voltage	3.7.5.1	4.7.5.1(D)	4.0%
Input voltage	3.7.5.4	4.7.5.4(D)	4.0%

(D) As in table I.

4.5.2.1 Group B sampling plan. The group B AQL shall be as shown in table III and the inspection level shall be S-4.

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in the order listed in table III.

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed group A and group B inspection. The inspection shall consist of the inspections specified in table IV. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for Group C inspections . An additional unit is required to perform operational checks during environmental testing.

4.5.3.1.1 Subgroup 1. For this subgroup, one unit shall be selected from the first production lot. For subsequent group C inspection, one unit shall be selected every third month. The unit shall be subjected to high and low temperature testing in that order.

4.5.3.1.2 Subgroup 2. For this subgroup, two units shall be selected from the first production lot. For subsequent group C inspection, one unit shall be selected every fourth month and one unit shall be selected every eighth month. The unit selected

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every fourth month shall be subjected to immersion, vibration, shock and immersion testing with the unit subjected to testing in that order. The unit selected every eighth month shall be subjected to EMI testing.

4.5.3.1.3 Subgroup 3. For this subgroup, one unit shall be selected from the first production lot. For subsequent group C inspection, one unit shall be selected every fifth month. The unit shall be subjected to humidity and rain testing.

4.5.3.1.4 Subgroup 4. For this subgroup, two units shall be selected from the first production lot. For subsequent group C inspections, two units shall be selected every twelfth month. One unit shall be subjected to altitude, dust and salt fog testing and one unit shall be subjected to fungus testing.

4.5.3.1.5 Subgroup 5. For this subgroup, one unit shall be selected from the first production lot. For subsequent group C inspection, one unit shall be selected every fourth month. The unit shall be subjected to the electrical tests in the order shown in table IV.

4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed for each subgroup in the order specified in 4.5.3.1.

4.5.3.3 Group C failures. Actions required relative to group C failures shall be as specified in the contract or purchase order.

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TABLE IV. Group C inspections.

Inspection	Reqt Para	Insp Para
Subgroup 1.		
High temperature	3.8.1.1	4.8.1.1
Low temperature	3.8.1.2	4.8.1.2
Subgroup 2.		
Immersion	3.8.7	4.8.7
Vibration	3.8.8	4.8.8.1
Bench handling	3.8.8	4.8.8.2
Transit drop	3.8.8	4.8.8.3
Immersion	3.8.7	4.8.7
EMI	3.9	4.9
Subgroup 3.		
Humidity	3.8.2	4.8.2(Note 1)
Rain	3.8.9	4.8.9
Subgroup 4.		
Altitude	3.8.3	4.8.3
Dust	3.8.4	4.8.4(Note 1)
Salt fog	3.8.5	4.8.5(Note 1)
Fungus	3.8.6	4.8.6(Note 1)
Subgroup 5.		
Output rise time	3.7.1.2	4.7.1.2
Output impedance	3.7.1.3	4.7.1.3
Input impedance	3.7.2.1	4.7.2.1
Receiver dynamic range	3.7.3.1	4.7.3.1
Input load impedance	3.7.3.2	4.7.3.2
Timing modes	3.7.3.3	4.7.3.3
Orderwire input impedance	3.7.4.1	4.7.4.1
Orderwire output impedance	3.7.4.2	4.7.4.2
Orderwire receiver level	3.7.4.3	4.7.4.3
Output waveshape	3.7.5.2	4.7.5.2(D)
Input and output impedance	3.7.5.3	4.7.5.3(D)
Analog channel input/ output impedance	3.7.7.1	4.7.7.1(A)

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TABLE IV. Group C inspections (continued)

Inspection	Reqt Para	Insp Para
Longitudinal balance	3.7.7.2	4.7.7.2(A)
Loop power feed (surge current only)	3.7.6	4.7.6(D)
Frame acquisition time	3.7.8.2	4.7.8.2
Detection of out-of-sync	3.7.8.4	4.7.8.4
Prime power	3.7.9	4.7.9(D)
Interchangeability	3.11	4.11

(D),(A) As in table I.

NOTE 1. The equipment shall be thoroughly washed, cleaned, dried and refurbished if necessary, before proceeding with subsequent tests.

4.5.4 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed group C inspection may be accepted on the contract provided all damage is reworked to specification requirements and the sample units are resubjected to and pass group A inspection only.

4.5.5 Group D inspection. This inspection shall consist of the tests specified in table V and shall be performed on units from lots which have been subjected to and have met group A and group B inspection. Test conditions, methods, and procedures shall be in accordance with 4.10.

TABLE V. Group D inspection.

Inspection	Reqt Para	Insp Para
Reliability	3.10	4.10

4.5.5.1 Sampling for Group D inspection. Eight units shall be selected at random from the first month's production and subjected to the test of 4.10. For subsequent group D inspections, eight units shall be selected at random from every fourth month's production.

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4.5.5.2 Group D failures. Actions required relative to group D failures shall be as specified in the contract or purchase order.

4.5.5.3 Reinspection of conforming group D sample units. Unless otherwise specified, sample units which have been subjected to and have passed group D inspection may be accepted on the contract provided all damage is reworked to specification requirements and the sample units are resubjected to and pass group A inspection only.

4.6 Electrical preconditioning (burn-in). (See 3.6) During electrical preconditioning, the equipment shall be connected per figures 2 and 3. The test conditions of 4.10.3 shall apply except that the duty cycle shall be 100%. If a failure occurs during this preconditioning, the failure shall be recorded, the failed component(s) replaced, and the preconditioning cycle reinstituted. Failure data obtained shall be submitted with the lot containing the unit in which the failure occured. The requirements of 3.6 shall be met.

a. Set all UUTs for 32 kb/s Channel Rates, even numbered UUTs to LOOP timing and odd numbered UUTs to MASTER timing. Set Timing Source to rate 3 (144 kb/s).

b. Connect the DSVT simulator to Loop Modem Channels 3 and 4 via the Power Feed Load Box and turn on all Loop Modem Power Feed switches. Do not connect any AAU channels.

c. Means shall be provided to latch all S, and SA alarms.

d. Turn power ON to all UUTs. Run the units for 48 hours, using BITE to indicate failure. If a failure is detected, it is to be repaired and recorded. The last 24 hours of burn-in, must be failure free.

e. After completion of burn-in, perform the operational test procedure outlined in 4.7.12.

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4.7 Electrical. (See 3.7) Unless otherwise specified all tests shall be performed at 115V ±5% from a single-phase source operating between 57.0 and 63.0 Hz. Tests shall be conducted at room ambient temperature, pressure, and humidity per MIL-STD-810 unless otherwise specified. The electrical test equipment listed in table VI or their equivalent will be utilized during testing and will be calibrated in accordance with MIL-STD-45662.The standard test equipment shall be provided by the contractor. Government technical representative approval is required prior to use of proposed substitute items. Unless otherwise specified, special test equipment (see table VII and figure 4) shall be provided by the contractor subject to approval by the Government. The Government shall be responsible for supplying the Government furnished equipment (GFE) required for test (see table VIII). Table IX identifies the external interface connections with the test configuration diagrams. Unless otherwise noted, all resistors shown in the figures are 0.5 watt.

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TABLE VI Standard test equipment.

Description	Mfg/Model No. (or equivalent)
Oscilloscope	HP 1740A
Wattmeter	Weston 432
Ammeter, dc	Weston 931 (3,15, and 30 amps)
Voltmeter, Digital	HP 3440A
Universal Counter	HP 5315A
Frequency Counter	HP 5328A
Squarewave/Pulse Generator	IEC P12
Voltmeter, ac	HP 403
Impedance Bridge	ESI 290A
Supply, DC 0-32 volts	Sorensen QSB28-8
Supply, AC Variable Frequency	Elgar 1001
Supply, DC 0-200 volts	Repco 605
Audio Oscillator	HP 200CD
Attenuator, 50 ohm (3 required)	HP 355C
Supply, DC 0-10V	Trygon Hr 40-500
Milliammeter, dc	Weston 931 (5, 20, 100, and 500 milliamperes)
True RMS Voltmeter	Ballantine Model 323
Distortion Meter	HP 339A
Envelope Delay Test Set	HP 4942A
Noise Measurement Set	HP 3555B

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Table VII. Special test equipment.

Equipment Name	Drawing No.	Functional Description
Group Simulator	A-3051667	The Group Simulator simulates the group data, orderwire and telemetry signals. It generates either a long or short pseudo-random sequence. The three signals can be generated simultaneously or in any combination. A TTL error output signal is provided for counting errors externally.
Timing Source		The Timing Source schematic is shown in figure 4.
DSVT Simulator	A-3051668	The DSVT Simulator provides a pseudo-random data generator and a correlator at 2, 16 and 32 kb/s with a balanced signal interface. A conditioned diphase interface is also provided at 16 and 32 kb/s. A TTL error output signal is provided for counting errors externally.
Error Inserter	A-3051665	The Error Inserter provides a means of injecting a.001 to .1% continuous BER with 20% random bursts. Balanced data and timing input and output interfaces are provided.

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Table VIII Government furnished equipment (GFE)

Equipment	Description
Alternate (for DSVT Simulator): Generator, Digital Data SG-1139/G	As an alternate for the DSVT Simulator, the Digital Data Generator provides pseudo-random data patterns of 72 to 4,608 kb/s balanced signals and 16/32 kb/s condtioned diphase signals. It provides correlation of digital data to perform bit-error measurements.
Modem, Digital Data MD-1026(P)/G	The Digital Data Modem provides electrical signal compatibility between balanced NRZ digital signals on its equipment side and conditioned diphase signals on its line side. This equipment must be configured with conditioned diphase CCAs in all four group slots.
Modem, Cable Driver MD-1024/G	The Cable Driver Modem provides high voltage cable power at 90 ma for repeatered cable transmission.
Cable Assembly CX-11230/G	Dual coax cable in 1/4 mile reel lengths.
Cable, Telephone WF-16/U	Four-wire twisted pair telephone cable.

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Table IX TD-1233 interface connections

Connector	Function	Pin LO	Pin HI	Pin GND	Signal Type
GROUP	Group Data In		*(M)		DIPHASE
	Group Data Out		*(F)		DIPHASE
CHANNELS 1-4	Data In Ch 1	B	b	L,q	DIPHASE
	Data Out Ch 1	F	X	L,q	DIPHASE
	Data In Ch 2	C	c	L.q	DIPHASE
	Data Out Ch 2	G	Y	L,q	DIPHASE
	Data In Ch 3	D	J	L,q	DIPHASE
	Data Out Ch 3	H	Z	L,q	DIPHASE
	Data In Ch 4	K	a	L,q	DIPHASE
	Data Out Ch 4	E	d	L,q	DIPHASE
STATION CLOCK AND ALARM	Station Clock In	A	B	C	SQ WV
	Alarm S	D	E	C	Alm=Close Contact
	Alarm SA	F	G	C	Alm=Close Contact
POWER	Power	C	A	B	
ORDERWIRE	AVOW In	A	B	C	
	AVOW Out	D	E	C	
	AVOW Control	F	G	C	Note 1

*(M) Male, (F) Female; Type detemined by center conductor

Note 1. These pins are connected together by the TS-3647 cable.

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4.7.1 Diphase group output . (See 3.7.1)

4.7.1.1 Diphase group output voltage. (See 3.7.1.1)

a. Connect the equipment as shown in figure 5.

b. Set the UUT switches as follows:

Loop Power Feed switches	OFF
Channel switch	32 kb/s
Timing switch	MASTER

c. Measure the group output peak-to-peak voltage (refer to figure 6).

d. Verify that the requirement of 3.7.1.1 is met.

4.7.1.2 Diphase group output rise and fall times. (See 3.7.1.2)

a. Connect the equipment as shown in figure 5.

b. Set the UUT switches as follows:

Loop Power Feed switches	OFF
Channel switch	32 kb/s
Timing switch	MASTER

c. Measure the group output rise and fall time (see figure 6).

d. Verify that the requirement of 3.7.1.2 is met.

4.7.1.3 Diphase group output impedance. (See 3.7.1.3)

a. Connect the UUT group output as shown in figure 7 with R1 =57.6 ohms.

b. Remove Timing Generator card. (Removal of Timing Generator CCA causes S alarm and turns off output data).

c. Set the UUT switches as follows:

Loop Power Feed switches	OFF
Timing switches	MASTER

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d. Set the oscillator to 1 kHz and its output voltage to 2.00 volts as measured with the voltmeter connected to V1 of the 57.6 ohm termination. Then measure V2.

e. Calculate the impedance using

$$Z = 57.6 \frac{V2}{V1-V2} \text{ ohms.}$$

Record the value.

f. Repeat d and e with the oscillator set to 300 kHz.

g. Connect the oscillator to Group Data Out through a 100 ohm, resistor, set it for 300 Hz and set its output voltage to 2.00 volts.

h. Calculate the impedance using

$$Z = 100 \frac{V2}{V1-V2} \text{ ohms.}$$

Record the value.

i. Repeat g and h with the oscillator set to 1 kHz.

j. Re-insert the Timing Generator CCA in UUT location A10.

k. Verify that the requirements of 3.7.1.3 are met.

4.7.2 Diphase group input. (See 3.7.2)

4.7.2.1 Diphase group input impedance. (See 3.7.2.1)

a. Connect the UUT Group input as shown in figure 7 with R1 = 57.6 ohms.

b. Set the oscillator to 1 kHz and its output voltage to 2.0 volts as measured with the voltmeter connected to V1. Then measure V2.

c. Calculate the impedance using

$$Z = 57.6 \frac{V2}{V1-V2} \text{ ohms.}$$

Record the value.

d. Repeat b and c with the oscillator set to 300 kHz.

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e. Connect the oscillator to Group Data Out through a 100 ohm resistor and set it for 300 Hz and its output voltage to 2.0 volts.

f. Calculate the impedance using

$$Z = 100 \frac{V2}{V1-V2} \text{ ohms.}$$

Record the value.

g. Repeat steps e and f with the oscillator set to 1 kHz.

h. Verify that the requirements of 3.7.2.1 are met.

4.7.2.2 Received signal level. (See 3.7.2.2)

a. Connect the equipment as shown in figure 8 and set the attenuator to 4 dB attenuation. Set oscillator to 1.6 kHz and +21.3 dBm.

b. Set the UUT switches as follows:

Loop Power Feed switches	OFF
Channel Rate switch	32 kb/s
Timing switch	MASTER

c. Measure the bit error rate on the first digital channel. Verify that there are no data errors.

d. Verify that the requirements of 3.7.2.2 are met.

e. Reduce the cable length to zero, eliminate the attenuator and loopback the Group signal.

f. Repeat steps c and d.

4.7.3 Station clock input. (See 3.7.3)

4.7.3.1 Receiver dynamic range. (See 3.7.3.1)

a. Connect the equipment as shown in figure 9.

b. Set the UUT switches as shown in 4.7.2.2, step b, with the timing source set to 144 kHz.

c. Using oscilloscope #2, adjust each attenuator to provide a ±50 mV peak signal referenced to ground.

d. Using oscilloscope #1, verify that the Station Clock input and Group Data output are in sync.

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4.7.3.2 Input load impedance. (See 3.7.3.2)

a. Connect the equipment as shown in figure 10.

b. Set the UUT as shown in 4.7.3.1.

c. Adjust the oscillator at a frequency of 1 kHz for an output level of 1 volt rms (V1). Then measure the V2 voltage.

d. Calculate the input impedance using

$$Z(in) = 100 \frac{V2}{V1} \text{ ohms.}$$

e. Verify that the requirements of 3.7.3.2 are met.

4.7.3.3 Timing modes. (See 3.7.3.3)

a. Loopback mode. Connect the equipment as shown in figure 11.

b. Set the UUT to 32 kb/s Channel Rate, Loopback Timing, Loop Power Feed switches OFF and Power ON.

c. Set frequency synthesizer to 144.0072 kHz.

d. Verify that Group Data In and Timing from I/O pin 23 of Timing Generator CCA A10 are in sync. Verify that there is no LED fault on CCA A10.

e. Set frequency synthesizer to 143.9928 kHz and repeat step d.

f. Master mode. Switch the UUT Timing to Master (up position) on CCA A11. Loopback to the group output. Measure the frequency at pin 23 of the Timing Generator CCA (A10). Verify that the frequency is within the range of 143.9942 kHz to 144.0050 kHz.

g. Station clock mode. This test is performed as part of the receiver sensitivity test as described in 4.7.3.1.

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4.7.4 Orderwire interface. (See 3.7.4)

4.7.4.1 Orderwire input impedance. (See 3.7.4.1)

a. Connect the AVOW In of the UUT as shown in figure 12.

b. Set the oscillator to approximately 2 Vrms at 1 kHz as measured at V1. Then measure V2.

c. Calculate the impedance using

$$Z\ (in) = 600\ \frac{V2}{V1}\ ohms.$$

d. Verify that the requirements of 3.7.4.1 are met.

4.7.4.2 Orderwire output impedance. (See 3.7.4.2)

a. Connect the AVOW Out of the UUT as shown in figure 12.

b. Set the oscillator to approximately 2 Vrms at 1 kHz as measured at V1. Then measure V2.

c. Calculate the impedance using

$$Z\ (out) = 600\ \frac{V2}{V1}\ ohms.$$

d. Verify that the requirements of 3.7.4.2 are met.

4.7.4.3 Orderwire receive level. (See 3.7.4.3) This test is performed as part of 4.7.12.2.

4.7.4.4 Bandwidth. (See 3.7.4.4)

a. Connect the UUT as shown in figure 13. Set the UUT for MASTER Timing and 32 kb/s Channel Rate.

b. Adjust the oscillator to 21.3 dBm at 1 kHz.

c. Measure and record the signal level appearing at the orderwire output using the digital voltmeter terminated in 600 ohms.

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d. Repeat steps b and c with the oscillator set at 300, 400, 3000 and 3400 Hz.

e. Verify that the requirements of 3.7.4.4 are met.

4.7.4.5 Signal-to-noise ratio. (See 3.7.4.5)

a. Connect the equipment as shown in figure 14.

b. Set the Auxiliary Unit and the UUT for 16 kb/s Channel Rate, and Loop Power Feed switches OFF. Set the Auxiliary Unit to MASTER Timing and the UUT to Loop Timing.

c. Set the DSVT Simulator for RANDOM data, Channel Rate to 16 kb/s, and Loop Timing.

d. Set the Transmission Test Set for 600 ohm termination and C-Message weighted.

e. Set the oscillator level so that the UUT OW output is -11.5 dBm at 1600 Hz. Measure the noise in dBm with the oscillator OFF.

f. Calculate the signal-to-noise ratio as follows:

$$SNR(dB) = V_{on}\ (dBm) - V_{off}\ (dBm).$$

g. Set the DSVT Simulator, Auxiliary Unit and UUT for 32 kb/s Channel Rate and repeat steps e and f.

h. Verify that the requirements of 3.7.4.5 are met.

4.7.5 Digital channel characteristics. (See 3.7.5)

4.7.5.1 Output voltage. (See 3.7.5.1)

a. Connect the equipment as shown in figure 15.

b. Set the DSVT simulator to 32 kb/s rate, Data to all zeros. Set the UUT to a 32 kb/s Channel Rate, MASTER Timing and all Loop Power Feeds OFF.

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c. Connect the DSVT Simulator to the first Digital Channel input. Measure the peak-to-peak channel output voltage as shown in figure 4.

d. Repeat c for each remaining Digital Channel.

e. Verify that the requirements of 3.7.5.1 are met.

4.7.5.2 Output waveshape. (See 3.7.5.2)

a. Connect the equipment as shown in figure 15.

b. Set the DSVT Simulator to 16 kb/s rate, all zeroes. Set the UUT to 16 kb/s Channel Rate, MASTER Timing and all Power Feeds to OFF.

c. Connect the DSVT Simulator to the first Digital Channel input. Measure the channel output rise and fall times and droop. Refer to figure 6 for a definition of these charteristics.

d. Repeat b and c for each remaining Digital Channel.

e. Verify that the requirements of 3.7.5.2 are met.

4.7.5.3 Input and output impedance. (See 3.7.5.3)

a. Connect the equipment as shown in figure 16.

b. Set the UUT for 32 kb/s Channel Rate, MASTER Timing, and all Power Feeds to OFF.

c. With the input setup connected to the first Digital Channel, adjust the oscillator for 1 volt, at a frequency of 8 kHz (V1). Measure V2.

d. Calculate the input impedance using

$$Z\ (in) = 125\ \frac{V2}{V1}\ \text{ohms.}$$

e. Repeat c and d with the oscillator frequency set to 16, 32 and 56 kHz.

f. Repeat c through e for each remaining Digital Channel.

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g. Verify that the input requirements of 3.7.5.3 are met.

h. With the output setup connected to the first Digital Channel output, measure and record the output voltage V1 with the switch open, then measure and record the voltage V2 with the switch closed.

i. Calculate the impedance using:

$$Z \text{ (out)} = 125 \left(\frac{V1}{V2}-1\right) \text{ ohms.}$$

j. Repeat h and i for each remaining Digital Channel.

k. Verify that the requirements of 3.7.5.3 are met.

4.7.5.4 Input voltage. (See 3.7.5.4)

a. Connect the equipment as shown in figure 17 and 18.

b. Set the DSVT Simulator for 32 kb/s operation and adjust the Line Simulator (switch to LOW) for 0.150 V peak-to-peak signal at the first Digital Channel input binding posts. Set the UUT for 32 kb/s Channel Rate, MASTER Timing, and Loop Power Feeds OFF.

c. Monitor the DSVT Simulator Error Detector.

d. Verify that the requirements of 3.7.5.4 are met.

e. Repeat b through d for each Digital Channel 2 through 4.

f. Set the Line Simulator switch to HIGH and adjust to 4.0 V peak-to-peak signal at the first Digital Channel input binding posts.

g. Monitor the DSVT Simulator Error Detector.

h. Verify that the requirements of 3.7.5.4 are met.

i. Repeat f through h for each Digital Channel 2 through 4.

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4.7.5.5 Idle loop. (See 3.7.5.5)

a. Connect the equipment as shown in figure 19, with the DSVT Simulator connected to Channel 1 of the UUT with S1 closed.

b. Set the UUT to 32 kb/s Channel Rate, MASTER Timing and Loop Power Feeds OFF. Set the DSVT Simulator to 32 kb/s Channel Rate.

c. Verify that there are no errors for 30 seconds.

d. Disconnect Channel 1 input to the UUT and verify that the requirements of 3.7.5.5 are met.

e. Repeat steps c and d for Channels 2 through 4.

4.7.6 Loop power feed. (See 3.7.6)

a. Connect the Loop Power Feed Load Circuit to the first Digital Channel as shown in figure 20.

b. Set S1 to A and enable the first Digital Channel Power Feed switch on the UUT. Then measure and record the current.

c. Set S1 to B and measure the voltage. Verify that the voltage is less than 56 V.

d. Set S1 to A and momentarily short either of the data output lines to either data input line. Then short each data output and data input momentarily to ground.

e. Verify that the current returns to at least 34.9 mA.

f. Turn OFF the Power Feed switch on the UUT.

g. Connect the loop power surge test setup as shown in figure 21 to the first Digital Channel.

h. Turn ON the Power Feed switch and adjust the pulse generator pulse width to 45 $\pm$1 ms and a repetition rate of 1 MHz. Verify that the loop power feed does not trip. Adjust the voltage V2 to a minimum of 1.0 volt, using the oscilloscope.

i. Repeat steps a through h for each remaining Digital Channel.

j. Verify that the requirements of 3.7.6 are met.

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4.7.7 Analog channel characteristics. (See 3.7.7)

4.7.7.1 Input/output impedance and return loss. (See 3.7.7.1)

a. Connect the equipment as shown in figure 22. Do not connect the setup to the UUT.

b. Set the oscillator to 300 Hz and adjust the output for a V1 voltage of 0 dBm. Record the V2 (set) voltage in dBm (Z test = ∞).

c. Connect the setup to the first Analog Channel input, reset the V1 voltage to 0 dBm and record the V2 (measured) voltage in dBm.

d. Repeat steps a through c for each remaining Analog Channel input.

e. Repeat steps a through d with oscillator frequency set to 3400 Hz.

f. Repeat steps a through e measuring each Analog Channel output.

g. Calculate the return loss, V2 (measured) minus V2 (set), in dB, and verify that the requirements of 3.7.7.1 are met.

4.7.7.2 Longitudinal balance. (See 3.7.7.2)

a. Set up the bridge as shown in figure 23.

b. Check the bridge balance at 300 and 3400 Hz without connecting the UUT. It should be at least 60 dB.

c. Connect the bridge to the first Analog Channel input.

d. Set V1 to 1 Vrms at 300 Hz. Then measure and record the V2 voltage.

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e. Repeat step d with the frequency set to 1000 Hz and 3400 Hz.

f. Repeat steps d and e with the bridge connected to the first Analog Channel output.

g. Repeat steps c through f for each of the remaining Analog Channels.

h. Calculate the longitudinal balance (LB) for each Analog Channel using the formula

$$LB = 20 \log \frac{V1}{V2}.$$

i. Verify that the requirements of 3.7.7.2 are met

4.7.7.3 Frequency response. (See 3.7.7.3) Frequency response is demonstrated as part of the individual CCA TRS tests performed on 100% of the analog CCAs.

4.7.7.4 Analog channel input and output test tone level and frequency. (see 3.7.7.4) The test tone level and frequency is demonstrated as part of the individual CCA TRS tests performed on 100% at the analog CCAs.

4.7.7.5 Input-output linearity. (See 3.7.7.5) Input-output linearity is demonstrated as part of the individual CCA TRS tests performed on 100% of the CCAs.

4.7.7.6 Loopback switch. (See 3.7.7.6) Operation of the loopback switch is demonstrated in 4.7.7.2 (normal) and 4.7.7.8 (loop).

4.7.7.7 Noise. (See 3.7.7.7)

4.7.7.7.1 Idle channel noise. (See 3.7.7.7.1)

a. Connect the equipment to the first Analog Channel as shown in figure 24.

b. Set the input and output level switches to 0 dB and the channel loopback switches to LOOP. Set the Channel Rate to 32 kb/s and the Timing switch to MASTER.

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c. Set the noise measuring set to 600 ohms terminated and C-message weighting.

d. Measure the channel output noise on channels 1 and 2.

e. Move the AAU card to the next channel CCA position and repeat steps c through e for channels 3 and 4.

f. Verify that the requirements of 3.7.7.7.1 are met.

4.7.7.7.2 Crosstalk. (See 3.7.7.7.2) For this test, the UUT must contain 4 Analog Channels.

a. Connect the equipment as shown in figure 25.

b. Set the UUT input level switches to -10 dB and the output level switches to 10 dB. Set Channel Loopback switches to NORMAL, Channel Rate to 32 kHz and MASTER Timing.

c. With the oscillator connected to Channel 1, adjust the oscillator at 820 Hz to produce an output voltage V1 of +6 dBm. (This gives a channel input of 0 dBm at 820 Hz).

d. With the transmission measuring set at 600 ohms termination, C message weighted, measure the V2 voltage at Channel 2 Out.

e. Repeat steps c and d with the transmission measuring set connected to Channel 3 and 4 outputs.

f. Verify that the requirements of 3.7.7.7.2 are met.

4.7.7.8 Peak clipping. (See 3.7.7.8)

a. Connect the equipment as shown in figure 26.

b. Set the Analog Channel Input Level switch to -10dB, and the Output Level switch to +10 dB. Set the Channel Loopback switch to LOOP, the channel rate switch to 32 kHz, and Master Timing.

c. With the oscillator connected to the first Analog Channel input, set the oscillator output at 200 Hz for +10 dBm, (this gives a Channel input of +4 dBm).

d. Measure total distortion.

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e. Repeat a through d for each Analog Channel.

f. Verify that requirements of 3.7.7.8 are met.

4.7.7.9 Full load tone. (See 3.7.7.9) The full load tone is demonstrated as part of the individual CCA TRS tests performed on 100% of the analog CCAs.

4.7.7.10 Analog channel signal processing. (See 3.7.7.10)

4.7.7.10.1 Clocking rate. (See 3.7.7.10.1) Clocking rate is tested as part of 4.7.7.10.3.

4.7.7.10.2 Dynamic range. (See 3.7.7.10.2) The dynamic range is demonstrated as part of the individual CCA TRS tests performed on 100% of the analog CCAs.

4.7.7.10.3 Output level control. (See 3.7.7.10.3) The output level control is demonstrated as part of the individual CCA TRS tests performed on 100% of the analog CCAs.

4.7.7.10.4 Input level control. (See 3.7.7.10.4) The input level control is demonstrated as part of the individual CCA TRS tests performed on 100% of the analog CCAs.

4.7.7.11 Envelope delay distortion. (See 3.7.7.11)

a. Connect the equipment as shown in figure 27.

b. Set the Analog Channel and Output/Input Level switches to 0dB and the Channel Loopback switches to LOOPBACK. Set the channel rate to 32 kbps, Group Rate 5, and Timing switch to MASTER.

C. Measure the envelope delay distortion of the input and output filter at 800 Hz and at 3000 Hz.

d. Repeat a and b for the remaining Analog Channels.

e. Verify that the delay is less than 100 us for each measurement.

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4.7.8 Framing. (See 3.7.8)

4.7.8.1 Format. (See 3.7.8.1) The TD-1233 output group patterns and overhead channels will be verified by examination of the group output data displayed on the oscilloscope. Channel positions will be verified by sequentially applying a diphase all zero pattern. Then, observing that the corresponding bit position in the group signal responds to the zeros, the channels will be located. The telemetry subchannel bit positions will be verified by applying activity to the first overhead subchannel (the framing subchannel counts as zero). For this test, configure the UUT with 4 digital channels.

a. Connect the equipment as shown in figure 28.

b. Set the UUT for 32 kb/s Channel Rate, and LOOP Timing.

c. Set the Group Simulator for

Group Rate	3
Timing Mode	INT
Transmit Format	MSF
Type	2
Channel Rate	32 kb/s
Data Select	Zeros

d. Observe the data displayed on the oscilloscope and locate the Framing Bit.

e. Connect the DSVT Simulator to Channel 1 and loopback Channels 2,3 and 4.

f. Verify that the group format is as shown in figure 29.

g. Repeat steps d and e for Channels 2, 3, and 4.

h. Set the Group Simulator to Subchannel 5. Verify that there are no errors for a 30 second period. Set the DSVT Simulator data to all ones.

i. Using the oscilloscope, verify activity on Subchannel 5 and that Subchannels 6 and 7 are digital ones.

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j. Repeat steps d through h with the Group Simulator Channel Rate set to 16 kb/s, Group Rate set to 1 and DSVT set to 16 kb/s.

k. Verify that the requirements of 3.7.8.1 are met.

4.7.8.2 Frame acquisition time. (See 3.7.8.2)

a. Connect the equipment as shown in figure 30. Configure both equipments for 32 kb/s Channel Rate. Set the UUT to MASTER Timing and the Auxiliary Unit to LOOPED Timing. Set the Group Modems to Rate 3, MOW.

b. Set up the error inserter for 0.1% BER continuous operation.

c. Squelch the data at the error inserter to cause a loss of frame; release squelch to restore data.

d. Measure the time from the resumption of received group data on error inserter to the resumption of data, at I/O pin 48 on the group framing card (CCA A9). Use the universal counter, take 50 measurements and verify that no more than 5 of the 50 measurements exceed 50 ms at 32 kb/s or 100 ms at 16 kb/s.

e. Set the error inserter for 20% BER random bursts interspersed with 0.1% BER ramdom error intervals.

f. Repeat step d, verify that no more than 5 of 50 measurements exceed 100 ms at 32 kb/s or 150 ms at 16 kb/s.

g. Set Group Modem to Rate 1, MOW and configure both equipments for 16 kb/s Channel Rate.

h. Repeat steps b through f.

4.7.8.3 Frame maintenance. (See 3.7.8.3)

a. The equipment setup is the same as shown in figure 30. Configure both equipments for 32 kb/s channel rate. Set the UUT to MASTER Timing, the Auxiliary equipment for LOOPED Timing, and the Group Modems to Rate 3, MOW.

b. Connect the event counter to TP6 on the group framing card(CCA A9) and adjust it to trigger on the positive edge of the signal. Squelch the data, this will cause the equipment to reframe. The counter should advance by one. Reset the counter.

c. Set the error inserter for 20% BER, continuous.

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d. Run the unit for a period of 12 hours. Verify that no declarations of Out-of-Sync are generated, as indicated by a count of zero on the counter.

e. Verify that the requirements of 3.7.8.3 are met.

4.7.8.4 Detection of Out-of-Sync. (See 3.7.8.4)

a. The equipment setup is the same as in 4.7.8.3.

b. Set the error inserter to no errors.

c. Use the data squelch signal (scope sync) at the error inserter to start the time interval measurement and use the signal at TP3 on the group framing card (CCA A9) to stop the counter.

d. Squelch the data, and measure the time from loss of data to out-of-frame indication.

e. Verify that no more than 5 of 50 measurements exceed 150 ms.

4.7.8.5 Idle filler pattern. (See 3.7.8.5)

a. Connect equipment as shown in figure 19 with the DSVT Simulator connected to Channel 1.

b. Set UUT to 32 kb/s channels, MASTER Timing and Loop Power Feeds OFF. Set DSVT Simulator to 32 kb/s Channel Rate.

c. Close S1 and verify that there are no errors for 30 seconds.

d. Open S1 and verify on the oscilloscope that the channel output is all diphase ones (16 kHz square wave).

e. Repeat steps c and d for Channels 2 through 4.

4.7.9 Prime power interface. (See 3.7.9)

a. Connect the equipment as shown in figure 31.

b. Configure the UUT for 32 kb/s Channel Rate, and MASTER Timing.

c. Connect the Digital Channels to the Loop Power Feed Load Box, such that the DSVT input and output is looped through all four Digital Channels.

d. Turn on all the Digital Channel Power Feed switches.

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e. Connect 115 Vac, 60 Hz power source to the UUT power connector.

f. Measure the input power.

g. Monitor Digital Channel errors over a 30 second period.

h. Repeat steps f and g for each ac input voltage and frequency shown in table X.

i. Connect 22 Vdc power supply to UUT and repeat steps f and g for each dc voltage shown in table X.

j. Disconnect power from the UUT power connector and connect high voltage dc power to the group connector for Power Feed measurements.

k. Repeat steps f and g for 180 and 196 Vdc power feed voltages.

l. Verify that the requirements of 3.7.9 are met.

Table X. Prime power interface input voltage and frequency combinations

Input Voltage	Frequency
115 Vac	60 Hz
103.5 Vac	60 Hz
126.5 Vac	60 Hz
103.5 Vac	47.5 Hz
103.5 Vac	420 Hz
126.5 Vac	47.5 Hz
126.5 Vac	420 Hz
22 Vdc	
28 Vdc	
32 Vdc	

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4.7.10 Current protection. (See 3.7.10)

a. Connect the equipment as shown in figure 32.

b. Set the UUT Remote Power switch to OFF.

c. With the UUT ac power turned OFF connect the MD-1024 Transmit output to the Group Data input of the UUT.

d. Turn on the MD-1024 unit power and Power Feed for 1 minute.

e. Verify that there is no physical damage to the UUT.

f. Repeat steps c through e for the Group Data output.

g. Loopback Group Data Out to Group Data In on the UUT and verify that there are no errors for 30 seconds on the DSVT Simulator.

4.7.11 BITE and alarms. (See 3.7.11 and 3.7.11.1)

a. With the UUT connected to a 115 Vac power source only, connect the equipment as shown in figure 33.

b. Set the UUT Power Feed switches to OFF, and the Channel Rate switches to 32 kb/s and MASTER Timing. Do not connect the Analog Channel inputs.

c. Turn the unit ON. Verify that the following BITE conditions are met.

(1) S must stay open.
(2) SA must stay open.
(3) Summary Fault lamp must be OFF.
(4) Power Out-of-Tolerance (POT) lamp must be OFF.
(5) All CCA LEDs must be OFF.

d. Verify that there are no erros for 30 seconds on the channels.

e. Remove the BITE 2 CCA and repeat step d.

f. Replace the BITE 2 CCA.

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g. Press the TEST button on the UUT. Verify that the BITE requirements listed below are met.

(1) S must stay closed.
(2) SA must stay closed.
(3) Summary Fault lamp illuminates.
(4) POT is ON.
(5) BITE indicator on each of the CCAs illuminates in a sequential order during self-test sequence.
(6) All indicators are OFF at the completion of the self-test sequence.

h. With the UUT power switch ON, disconnect 115 Vac power source.

(1) S must stay closed.
(2) SA must stay closed.

i. Turn OFF the UUT power switch. Verify that the following BITE requirements are met.

(1) S must stay closed.
(2) SA must stay open.

4.7.12 Operational check (See 3.7.12 and 3.13)

a. Connect the equipment as shown in figure 33.

b. Set all Analog Channel Loopback switches to normal and Level switches to 0 (AAU CCAs). Set all Loop Power Feed switches ON (Loop Modem CCAs)

c. Set the UUT for 32 kb/s Channel Rate and MASTER Timing, and loopback the Group signal.

d. Set the Orderwire oscillator output to +21.3 dBm at a frequency of 1600 Hz.

e. Connect the UUT to 115 Vac power and turn the unit ON.

f. Perform the tests specified in 4.7.12.1 through 4.7.12.5.

4.7.12.1 Digital data error check. (See 3.7.12.1)

a. Connect the DSVT Simulator to the looped Digital Channels.

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b. Monitor the Error Detector for a 30 second period.

c. Verify that the requirements of 3.7.2.1 are met.

4.7.12.2 Maintenance OW output check. (See 3.7.12.2) Measure the OW output with the transmission measuring set. Verify that the requirements of 3.7.12.2 are met.

4.7.12.3 DC voltage. (See 3.7.12.3)

a. Measure the dc voltage on each Digital Line at the Power Feed Load Box.

b. Repeat step a with the unit connected to 28 Vdc power and 180 Vdc high voltage cable power.

c. Verify that the requirements of 3.7.12.3 are met.

4.7.12.4 Sync frame check. (See 3.7.12.4) Monitor the DSVT Error Detector while turning the unit off and on. When the power is turned ON there shall be no errors, indicating that the unit has reframed.

4.7.12.5 Analog channel check. (See 3.7.12.5)

a. Connect the Analog Channel input to the first Analog Channel through 600 ohms. Set the oscillator at 820 Hz for an output of -11 dBm.

b. Connect the Analog channel output to the voltmeter terminated in 600 ohms.

c. Measure the first Channel output in dBm. Verify that the requirements of 3.7.12.5 are met.

d. Repeat steps a through c for each remaining Analog Channel.

4.7.12.6 Alarm monitor check. (See 3.7.12.6)

a. Connect the UUT to 115 Vac and verify that there are no alarms generated.

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b. Remove loopback and verify that the unit generates an S and a LOI alarm and no SA alarm.

c. Reconnect the loopback and verify that there are no alarms.

d. Verify that the requirements of 3.7.12.6 are met.

4.7.13 Environmental operational test. (See 3.7.13)

a. For all tests except temperature, connect the equipment as shown in figure 34 or 35 depending upon unit configuration. For temperature tests, the unit must be configured as shown in figure 33.

b. Set the UUT (AAU CCAs) Analog Channel Loopback switches to NORMAL and Level switches to 0. Set UUT (Loop Modem CCAs) Power Feed switches on.

c. Set the UUT and Auxiliary Unit for 32 kb/s Channel Rate. Set the UUT to Looped Timing and the Auxiliary Unit to MASTER Timing.

d. Connect the DSVT Simulator to the Looped Digital Channel on the Auxiliary Unit.

e. For all tests except temperature, connect the UUT to 28 Vdc power.

4.7.13.1 Digital channel check. (See 3.7.13.1) Monitor the Digital Channel Error Detector for a 30 second period. Verify that there are no errors.

4.7.13.2 Maintenance OW ouput check. (See 3.7.13.2) With the oscillator set to 21.3 dBm at 1600 Hz, verify that the requirements of 3.7.13.2 are met.

4.7.13.3 DC voltage check. (See 3.7.13.3) Measure the dc voltage on each digital line at the Power Feed Load Box. Verify that the requirements of 3.7.13.3 are met.

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4.7.13.4 Sync frame check. (See 3.7.13.4) Monitor the UUT Summary Alarm while turning off the Auxiliary Unit. The UUT shall generate a Summary Fault and a LOI. Turn on the power to the Auxiliary Unit. The UUT Alarms shall extinguish, indicating that the unit has reframed.

4.7.13.5 Analog channel check. (See 3.7.13.5)

a. Connect the oscillator to the first Analog Channel input on the UUT. Set the oscillator at 820 Hz for an input of -11 dBm through 600 ohms.

b. Connect the Analog Channel output on the Auxiliary Unit to the voltmeter terminated in 600 ohms.

c. Measure the Channel output (UUT to Auxiliary Unit) in dBm.

d. Repeat steps a through c in the other direction (Auxiliary Unit to UUT).

e. Repeat steps a through d for each remaining Analog channel.

f. Verify that the requirements of 3.7.13.5 are met.

4.7.13.6 Alarm monitor check. (See 3.7.13.6)

a. Verify that there are no alarms generated.

b. Shut off the Auxiliary Unit. Verify that the UUT generates an S and a LOI alarm and no SA alarms.

c. Turn on power to the Auxiliary Unit. Verify that there are no alarms.

d. Verify that the requirements of 3.7.13.6 are met.

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4.8. Service conditions. (See 3.8)

4.8.1 Temperature. (See 3.8.1)

4.8.1.1 High temperature. (See 3.8.1.1) The equipment shall be subjected to the test of MIL-STD-810D, Method 501.2, Procedure II for cycling exposure. In step 12, raise the internal chamber temperature to +51.7°C (+125°). Maintain internal chamber temperature for six hours at +51.7°C (+125°F). Raise the internal chamber temperature to +71.1°C (+160°F) within a time period of one hour and maintain at that temperature for four additional hours. Lower the internal chamber temperature to +51.7°C (+125°F) within a time period of one hour. Repeat the above steps two additional times to make a total of three 12-hour cycles. Step 14 shall be performed as follows: adjust the internal chamber temperature to +71.1°C (+160°F) and maintain this temperature until temperature stabilization of the test item is reached. Operate the test item until the item is stabilized.

4.8.1.2 Low temperature. (See 3.7.1.2) The equipment shall be subjected to the test of MIL-STD-810D, Method 502.2, Procedure I. Lower the internal chamber temperature in step 2 to -57°C (-70°F) for a period of not less than two hours following stabilization of the equipment, or until 24 hours have elapsed, whichever occurs first. In step 5 of Procedure I, perform Procedure II - Operation (paragraph II-3.2 of Method 502.2). Chamber temperature in step 1 of Procedure II shall be adjusted to -45°C (-50°F).

4.8.2 Humidity. (See 3.8.2) The equipment shall be subjected to the test of MIL-STD-810D, Method 507.2, Procedure III. After completion of Procedure III, the test item shall be subjected to Procedure II. In step 4, cycle 5 of table 507.2-I shall be used.

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4.8.3 Altitude The equipment shall be subjected to the test of MIL-STD-810D, Method 500.2, Procedure I. For step 2, adjust the chamber pressure to 141 mm of Hg (5.56 inches of Hg or 40,000 feet above sea level), at a rate not to exceed 2,500 fpm. The equipment shall also be subjected to the test of MIL-STD-810D, Method 500.2, Procedure II. The chamber pressure in step 2 shall be adjusted to 429 mm ofHg (16.9 inches of Hg or 10,000 feet above sea level) at a rate not to exceed 2,500 fpm.

4.8.4 Dust. (See 3.8.3) The equipment shall be subjected to the test of MIL-STD-810D, Method 510.2, Procedure I. The air velocity in step 2 shall be 3542± 200 feet per minute. In step 5, reduce the air velocity to 300 ± 200 feet per minute and adjust the air temperature to + 63°C(+145°F). The equipment shall be reoriented periodically to equally expose all sides of the equipment to the dust stream. Equipment shall be operated before step 1, during step 8 and during step 11. There shall be no degradation in performance as a result of the dust test, and there shall be no binding or blocking of moving parts, non-operation of contacts or relays, clogging or dust accumulation which could form the nucleus for the collection of water vapor as a result of dust penetration of the test item.

4.8.5 Salt fog. (See 3.8.5) The equipment shall be subjected to the test of MIL-STD-810D, Method 509.2, Procedure I. In step 2, the 48 hour period shall be constant wetting. Delete step 3. Operation in step 4 shall take place after gentle wash in running water not warmer than +38°C (+100°F) and drying. There shall be no corrosion of finishes or metals. Such corrosion shall be defined as any visible degradation of the equipment surfaces that can be attributed to flaky, pitted, blistered or otherwise loosened finish on metal surfaces. There shall also be no clogging or binding of moving mechanical parts.

4.8.6 Fungus. (See 3.8.6) The equipment in the assembled-and-ready-for-delivery-condition, or representative samples subject to approval of the procuring agency, shall be subjected to the test of MIL-STD-810D, Method 508.3. After 28 days, based on visual examination, the equipment or samples shall show no more than sparse microbial growth with restricted tuberal growth development in an area 10% or less of the total area and no more than six unrelated minute colonies with mycelial development in non-critical areas. Critical areas include: terminal spacing, CCAs etc. The equipment shall fail if it shows more than the growth specified above.

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4.8.7 Immersion. (See 3.8.7) The equipment shall be subjected to the test of MIL-STD-810D, Method 512.2, Procedure I.

4.8.8 Vibration and shock. (See 3.8.8)

4.8.8.1 Vibration. The equipment shall be subjected to the test of MIL-STD-810D, Method 514.3 for Category 1 -Basic Transportation. The equipment in its transit case, if any, as restrained and/or tied down in its combat transport configuration shall be subjected to the random vibration environment for mission/field transportation. The vibration environment shall be simulated using the test schedules given in figures 514.3-4, 514.3-5, 514.3-6, and figures 514.3-10 through 514.3-22. Test duration for figures 514.3-4, 514.3-5, and 514.3-6 shall be 48 minutes per axis, for a test time of 144 minutes. Test duration for figures 514.3-10 through 514.3-22 shall be as given in the test schedules, for a test time of 3 hours. Total test time shall be 324 minutes. The test item shall not be operated during this test. The equipment in its transit case, if any, shall also be subjected to the test of MIL-STD-810D, Method 514.3 for transportation vibration, Category 3 -Loose Cargo Transport. Prior to and following the tests for both Category 1 and Category 3, the equipment shall be operated in accordance with MIL-STD-810D, General Test Performance Guidance, paragraph 4.5. The equipment shall not have any degradation in performance. The equipment, including transit case, shall not sustain any mechanical damage.

4.8.8.2 Transit drop. The equipment in its transit case, if any, shall be subjected to the test of MIL-STD-810D, Method 516.3, Procedure IV. Prior to and following this test, the equipment shall be operated in accordance with MIL-STD-810D, General Test Performance Guidance, paragraph 4.5. The equipment shall not have any degradation in performance. The equipment, including transit case, shall not sustain any mechanical damage.

4.8.8.3 Bench handling. The equipment shall be subjected to the test of MIL-STD-810D, Method 516.3, Procedure VI. Prior to and following this test, the equipment shall be operated in accordance with MIL-STD-810D, General Test Performance Guidance, paragraph 4.5. The equipment shall not have any degradation in performance and shall not sustain any mechanical damage.

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4.8.9 Rain. (See 3.8.9) The equipment shall be subjected to the test of MIL-STD-810D, Method 506.2, Procedure I. In step 1, the test item shall be exposed to a simulated rain at a rate of 2 ± 0.5 inches per hour for 10 minutes. The rate of rainfall shall then be raised to 5 ± 1 inches per hour and held at this rate for 5 minutes. The rate shall then be reduced to 2±0.5 inches per hour for the next 15 minutes. Step 2 shall be changed as follows: starting 5 minutes after the initiation of the rain, the wind source shall be turned on and adjusted to produce a horizontal wind velocity of 40 miles per hour (3,500 feet per minute). The test item shall be operated in step 3.

4.9 EMI tests. (See 3.9) The unit shall be tested for compliance with 3.9. Test methods and procedures shall be in accordance with MIL-STD-462, Notice 3, and a Government-approved test plan. During susceptibility testing, the integral test facility plug-in subassembly shall be used to monitor operation of the unit.

4.9.1 Bonds and grounds. (See 3.9.1) Prior to performance of EMI testing, compliance with requirements of 3.9.1 shall be ascertained. Bonding resistance measurements shall be recorded and included in the EMI test report.

4.10 Reliability testing. (See 3.10)

4.10.1 First article reliability test. First Article reliability testing shall be in accordance with Test Plan VIC, of MIL-STD-781C.

4.10.2 Quality conformance (Group D) reliability test. The reliability test for Group D inspection shall be in accordance with Test Plan VIC, of MIL-STD-781C.

4.10.3 Reliability test conditions. Reliability testing shall be accomplished subject to the following conditions:

a. Two equipments shall be operated back-to-back.

b. Duty cycle shall be as follows: Apply prime power to the UUT for 4 hours. Turn power OFF for 2 hours. Turn power ON for 16 hours. Turn power OFF for 2 hours. Repeat this sequence for each 24-hour cycle.

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c. Chamber temperature shall be 50°C to 55°C (122°F to 131°F).

d. Vibration shall be 2.2 g ± 10% peak acceleration value at any nonresonant frequency between 20 and 60 cps measured at the mounting points on the equipment. The duration of vibration shall be at least 10 minutes during each hour of equipment operating time.

4.10.4 Reliabilty test method. The detailed test proecedures, and conditions for performing the tests outlined in 4.10.1 and 4.10.2 shall be in accordance with a contractor-prepared, Government-approved test plan.

4.10.5 Failure definition. A failure is defined as any malfunction which causes or may cause inability to commence operation, cessation of operation, or degradation below limits specified in 3.7.12.

4.10.6 Failure reporting and analysis. A failure reporting and analysis system shall be implemented per MIL-STD-781. Final determination of the failure relevancy and multiple and pattern failure categorization shall be the responsibility of the procuring activity.

4.11 Interchangeability. (See 3.11) The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirements of 3.11. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of case, cover, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of switches, sockets, and fasteners.

d. Location of panel-mounting screw holes on basic unit for mating with case.

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e. Location of connectors, locking pins, fasteners, slides and mountings which receive mating parts or plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

4.12 Electrical subassembly tests. (See 3.12) Plug-in subassemblies, including spares, shall meet the requirements of their respective test requirements specification drawings listed in 3.12.

4.13 Visual and mechanical. (See 3.14 and 3.15) The equipment shall be examined for the defects listed in MIL-STD-252 to determine compliance with 3.15. An inspection shall be performed to verify compliance with those portions of 3.14 (System safety engineering) which can be determined visually.

4.13.1 System Safety Design Verification Check List. An inspection shall be performed, on one unit only, during first article testing, utilizing SEL Form 1183, to verify compliance with 3.14 (System Safety Engineering.)

4.14 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First article inspection of packaging.

b. Quality conformance inspection of packaging.

4.14.1 First article inspection of packaging. Unless otherwise specified in the contract, first article inspection of packaging shall be in accordance with the Unit Pack Design Validation Requirements of MIL-P-116.

4.14.2 Quality conformance inspection of packaging.

4.14.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

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4.14.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B Quality Conformance Inspection Requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.14.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in table XI, "Packing inspection provisions". Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and AQL of 4.0% defective.

TABLE XI. Packing inspection provisions.

NO.	CHARACTERISTIC	METHOD OF INSPECTION
101	Intermediate container not as specified	Visual
102	Improper closure of intermediate container	Visual
103	Shipping containers not in accordance with specification	Visual
104	Excessive cube	Visual
105	Improper blocking and bracing	Visual
106	Closure not in accordance with specification	Visual
107	Weight and size exceed container limitations	Weight & Measure
108	Strapping not in accordance with specification, incorrectly applied, omitted	Visual
109	Marking omitted, incorrect, or illegible	Visual

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4.15 Serial numbers. (See 3.5.1) The equipment shall be inspected to verify that the serial numbers appear on each nomenclatured item and CCA.

4.16 Nuclear tests. (See 3.16) Tests for nuclear survivability shall be preformed on the equipment in accordance with Section 3 of CR-CX-0173-001.

5. Packaging.

5.1 Preservation. Preservation shall be as specified in MIL-STD-2073-2, tables I through VII., coded as follows:

5.1.1 Level A. 10-1-00-00-NS-X-ED

5.1.2 Level B. Same as level A.

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES.

6.1 Intended use. The equipment is designed to be rugged and compact, suitable for portable operation in tactical communications field locations (exposed) and for use in shelters and vans. It operates with the Digital Secure Voice Terminal (DSVT) KY-68 and the Digital Non-secure Voice Terminal (DNVT) TA-954/TT as a loop input/output via Telephone Cable WF-16. The unit also provides power feed to the voice terminals. The group input/output is compatible with the Multiplexer-Combiner, TD-1234(P)/TCC, the Modem, Cable Driver, MD-1025/G and the Modem, Digital Data, MD-1026(P)/G.

6.1.1 Functional description. The equipment is designed to time division multiplex/demultiplex up to 4 analog or conditioned diphase loops into/from a 4-1/2 channel group. A half-rate channel is allocated to overhead information including a 2/4 kb/s framing subchannel and a 2 kb/s telemetry subchannel. The group

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input/output is a conditioned diphase signal suitable for operation over CX-11230 cable. The equipment is capable of being powered from the group cable. An analog voice orderwire is provided for setup and maintenance of the CX-11230 group cable system. The 2 kb/s telemetry channel is looped back.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level A or B preservation and packing (see Section 5).

c. When first article rough handling tests are not required.

d. Quantity of first article inspection units required.

e. Place of final inspection.

f. Technical literature required.

g. Quantity of tools and running spare parts required.

h. Group C failure requirements.

i. MIL-Q-9858 .

j. Length of cables (if any required).

k. When first article packaging inspection reports require Acquisition Activity approval prior to production unit packing.

6.3 Environmental. Environmental pollution prevention measures are contained in the packing material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

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6.4 Final inspection. (See 4.5.3) Approval to ship may be withheld at the discretion of the Government pending the decision from the contracting officer on the adequacy of corrective action.

6.4.1 Initial shipments. Initial shipments of end items will be contingent upon successful completion of all first article tests.

6.5 Location of operational inspection. It is desirable that the operational system test be performed at a location that will minimize handling after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.6 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contract's quality control system and the quality history of the product.

6.7 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage, and handling.

6.8 Government-furnished property. The contracting officer should arrange to furnish the property listed in table VIII.

6.9 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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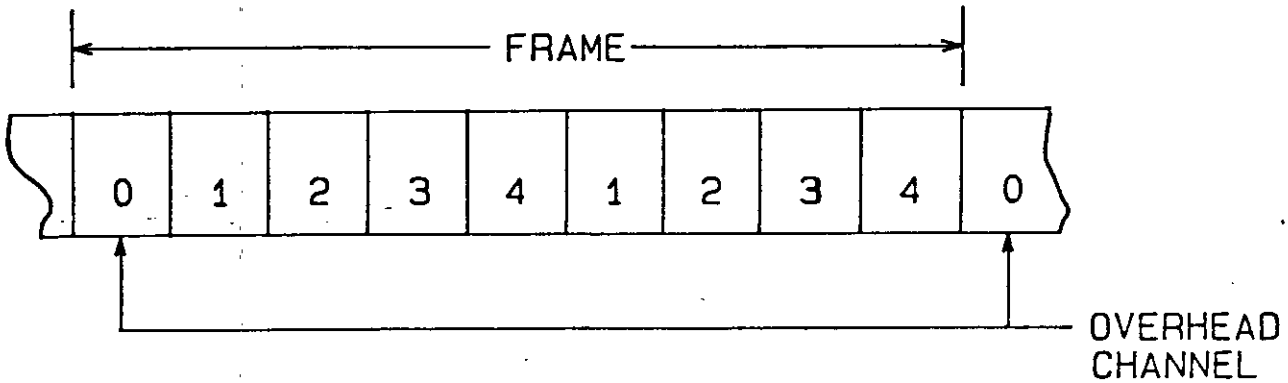
6.10 Subject term (key word) listing.

Analog Applique
Continuous Variable Slope Delta
CVSD
Digital Group Multiplex
Diphase
Framing
Group Modem
Loop Modem
Multiplexer
Multiplexer, TD-1233 (P)/TTC
MUX/DEMUX
Orderwire
Remote Loop Group Multiplexer (RLGM)

CUSTODIAN:
Army - CR

PREPARING ACTIVITY:
Army - CR
Project 5820-A829

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GROUP FORMAT

0	1	2	3
FRAME	TELEMETRY	SPARE SET TO 1	SPARE SET TO 1

OVERHEAD FORMAT

FIGURE 1. TD-1233 group and overhead format

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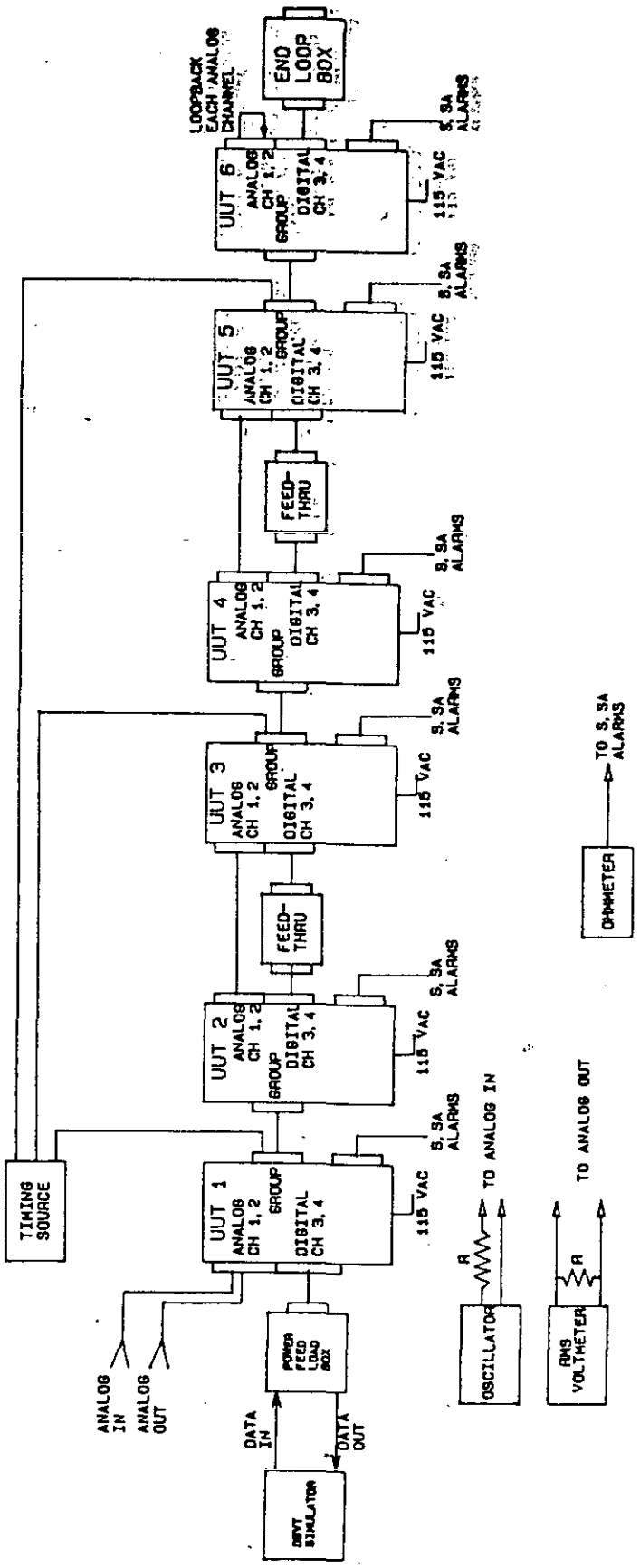


FIGURE 2. Burn-in.

FOR DETAIL OF TEST BOXES SEE FIGURE 3.

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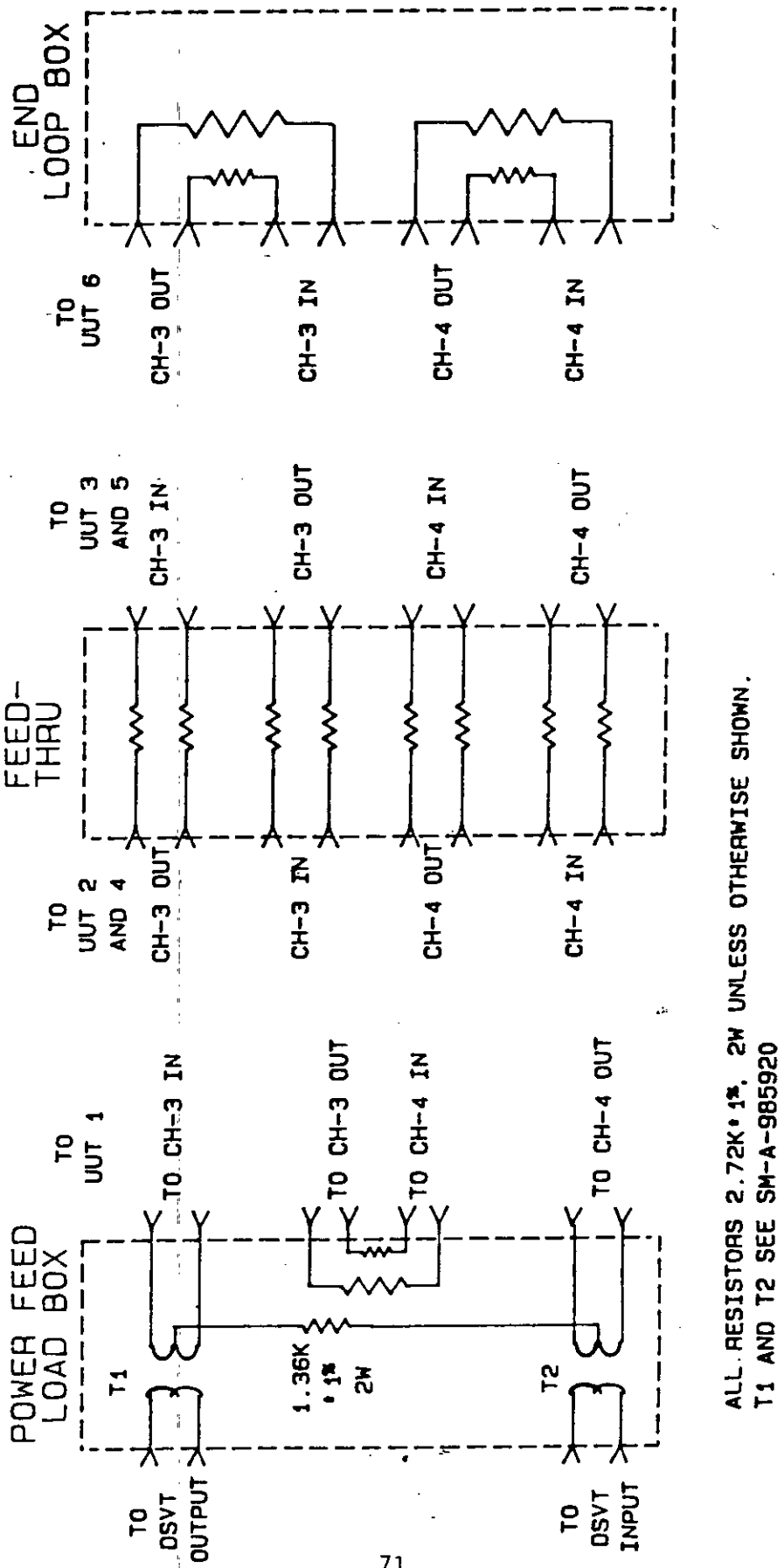
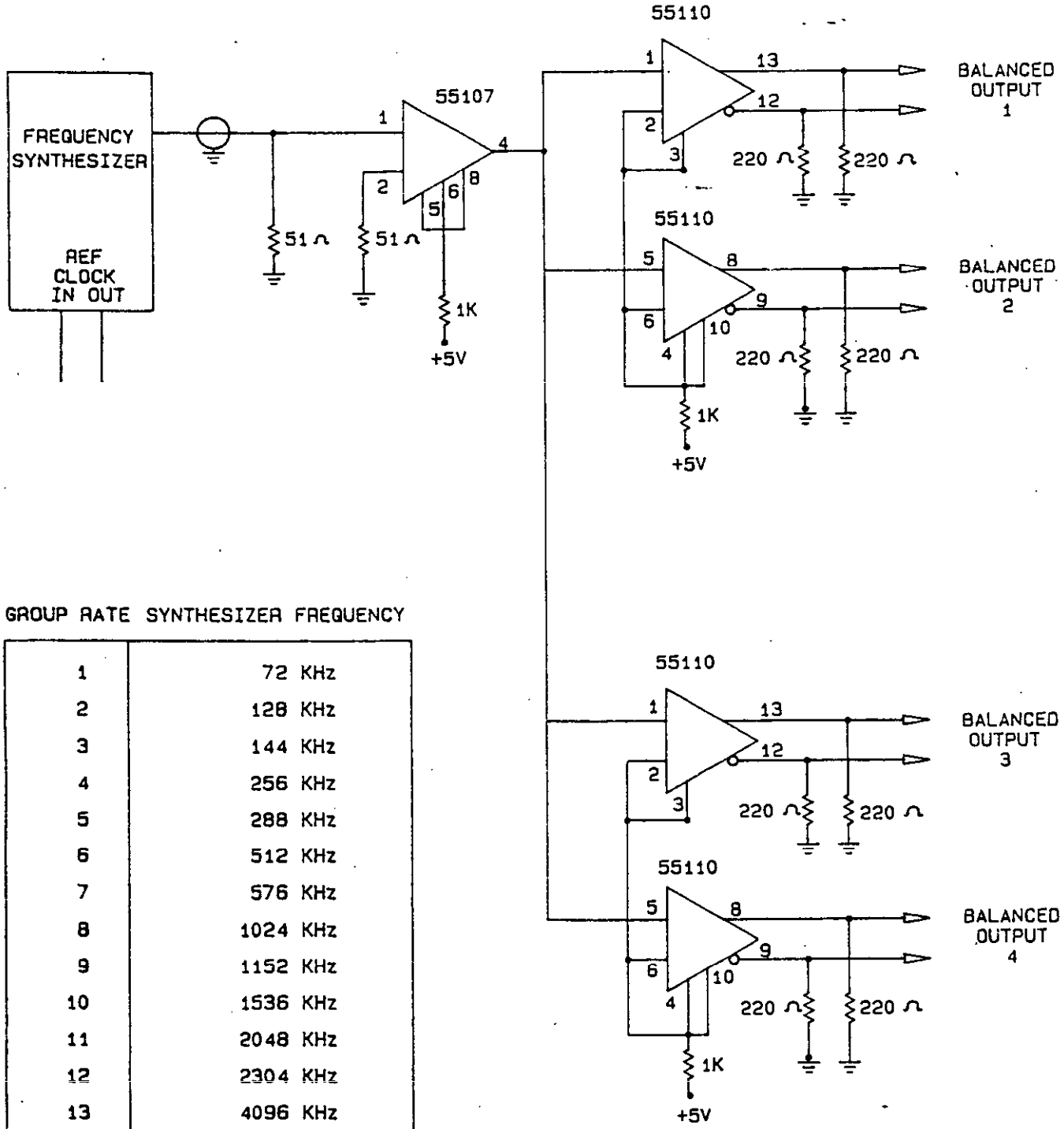


FIGURE 3. Burn-in test boxes.

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GROUP RATE SYNTHESIZER FREQUENCY

1	72 KHz
2	128 KHz
3	144 KHz
4	256 KHz
5	288 KHz
6	512 KHz
7	576 KHz
8	1024 KHz
9	1152 KHz
10	1536 KHz
11	2048 KHz
12	2304 KHz
13	4096 KHz
14	4608 KHz
15	4915.2 KHz

FIGURE 4. Timing source.

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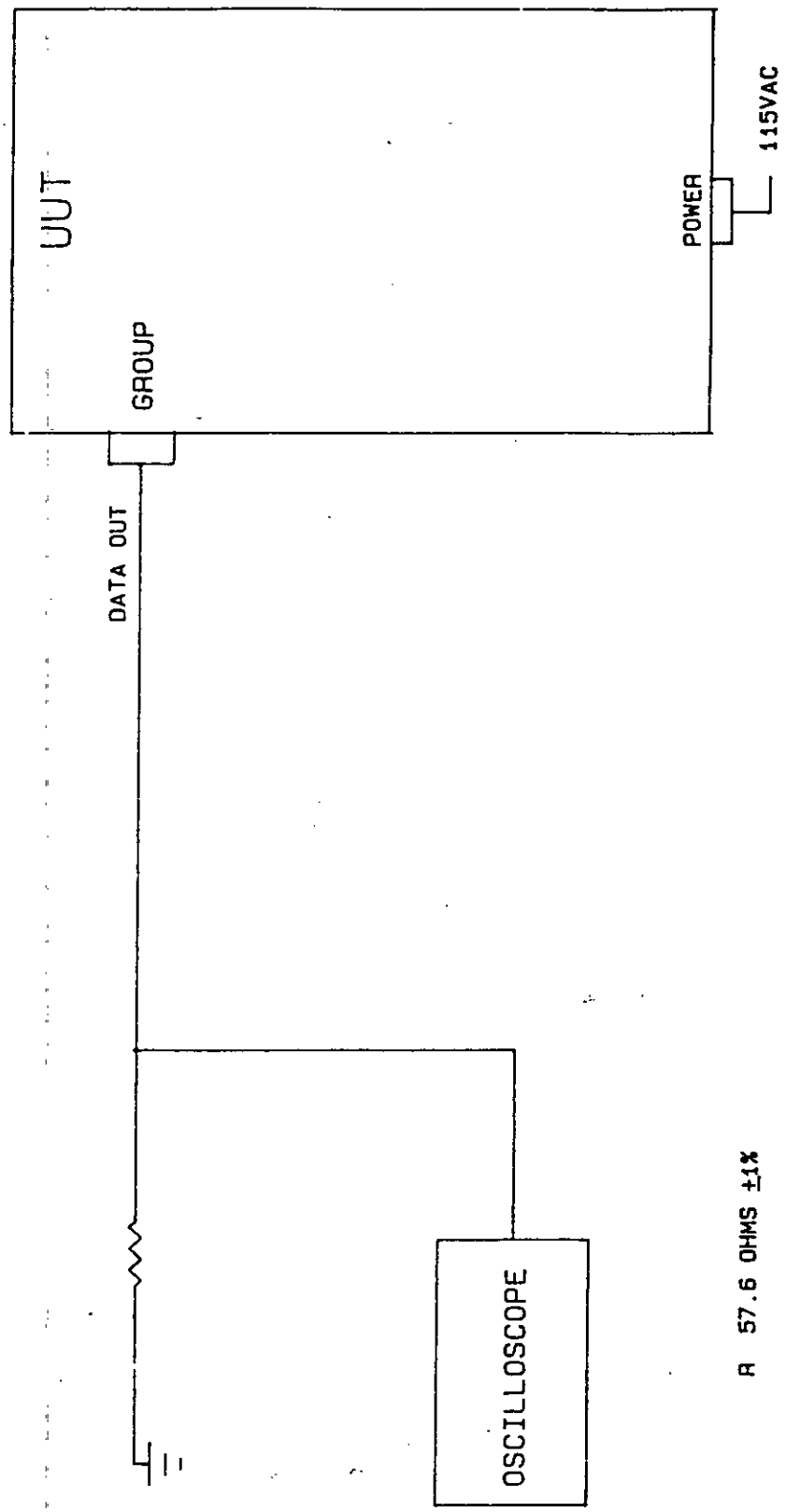


FIGURE 5. Diphase group output voltage and rise-time.

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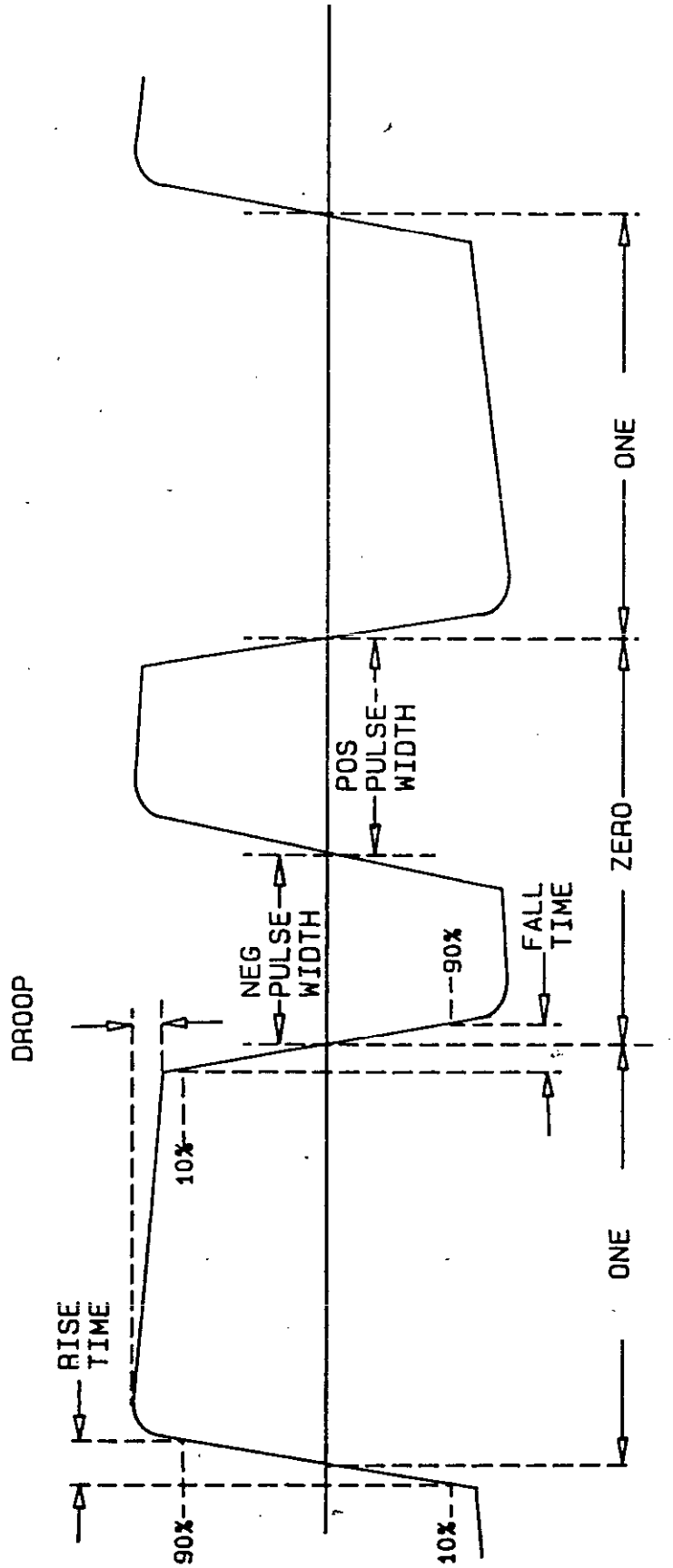


FIGURE 6. Diphase waveform (digital zero).

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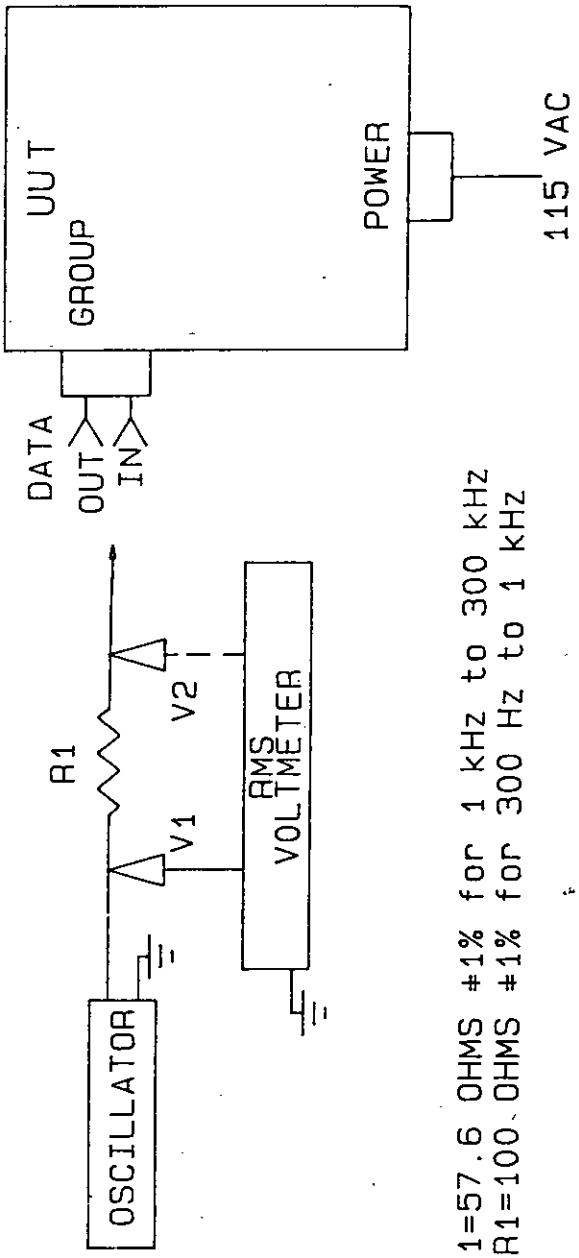


FIGURE 7. Diphase group output/input impedance.

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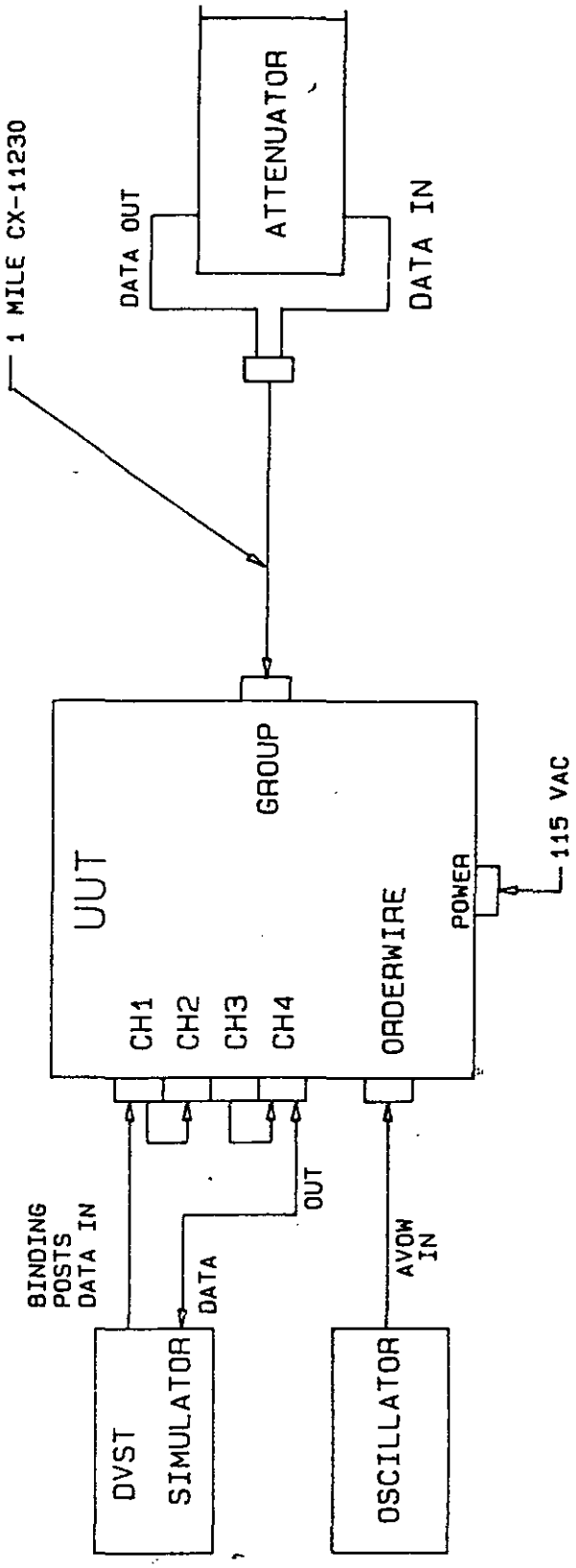


Figure 8. Received signal level test

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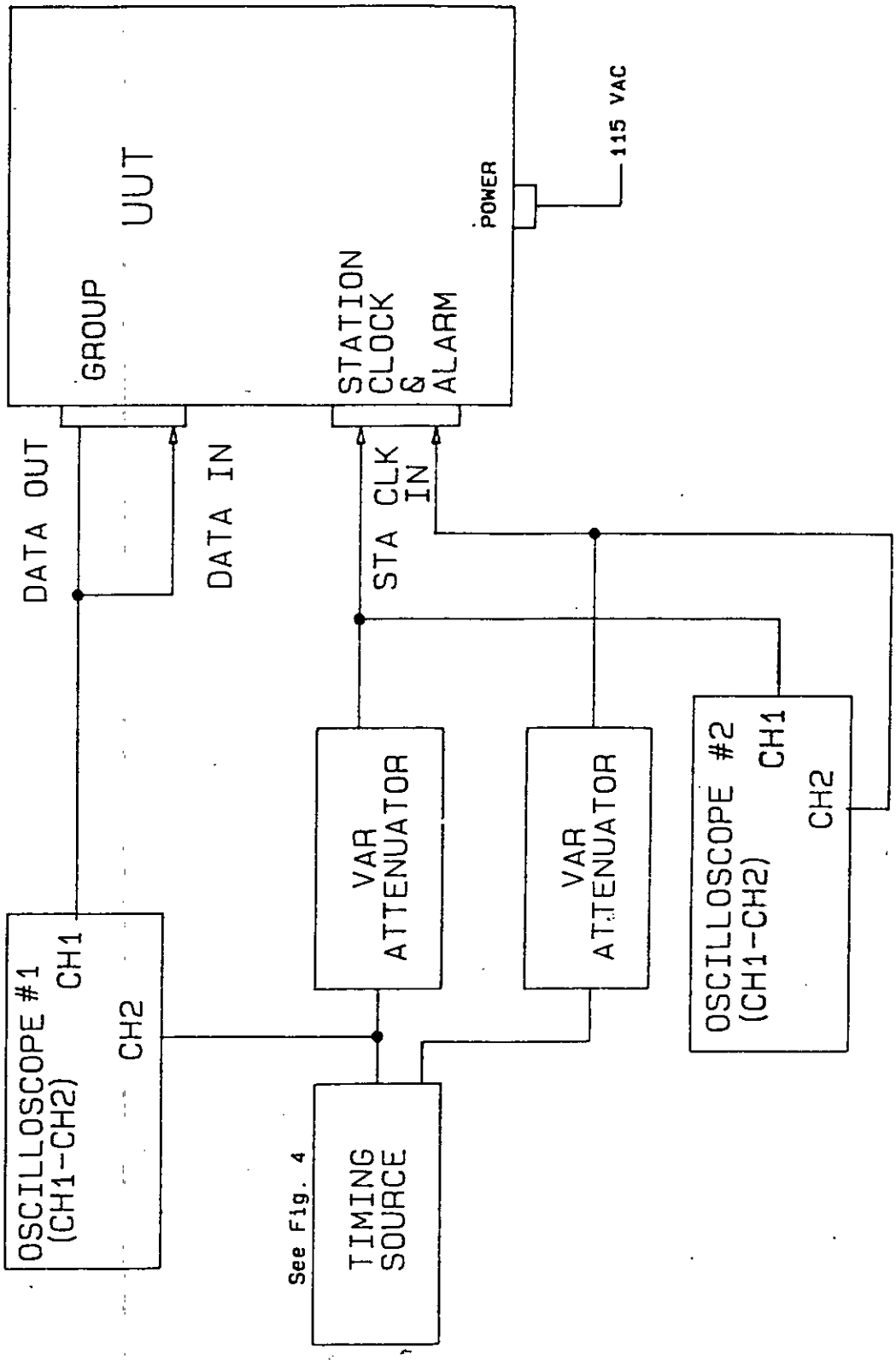


FIGURE 9. Receiver sensitivity and timing mode.

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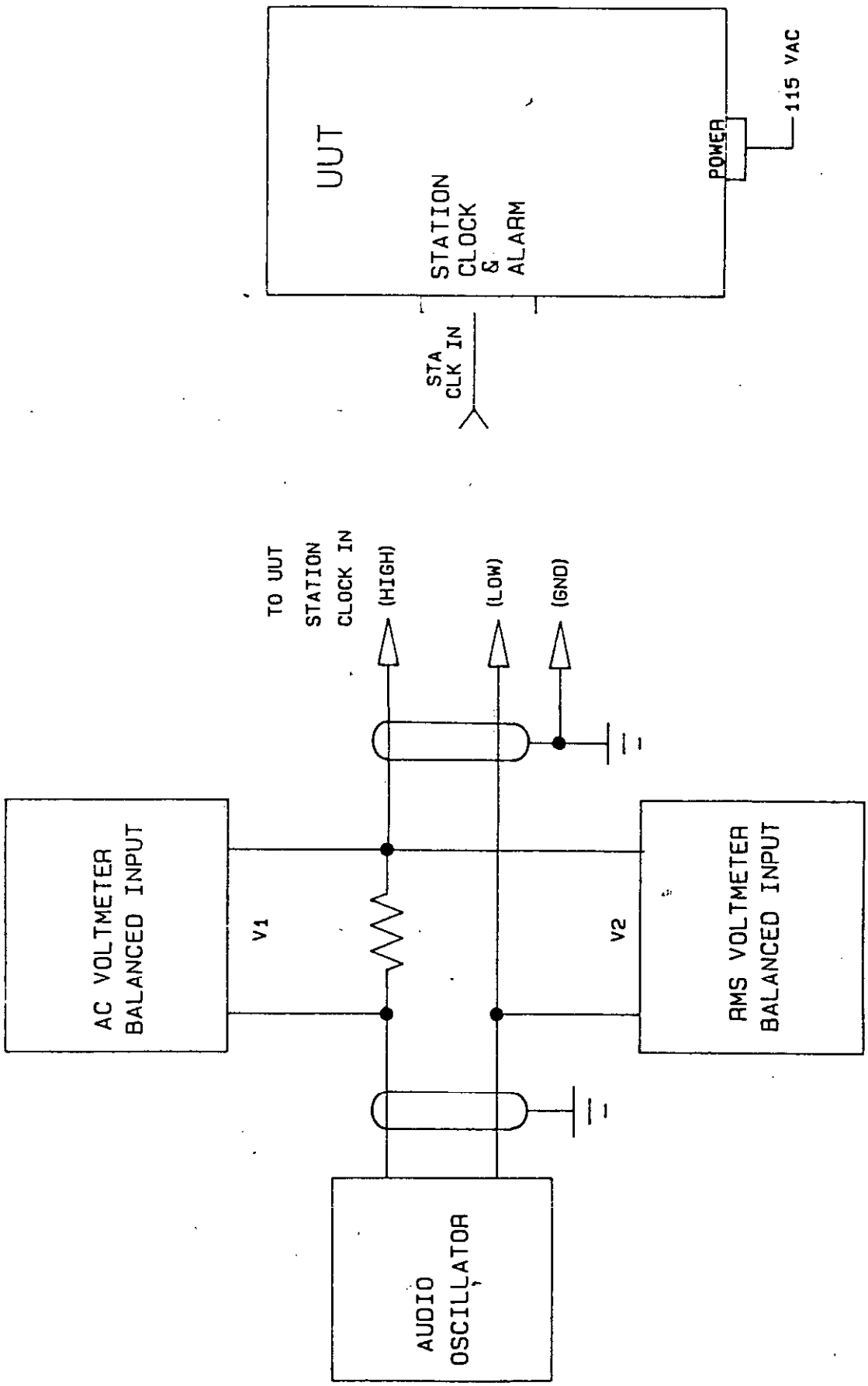


FIGURE 10. Input load impedance.

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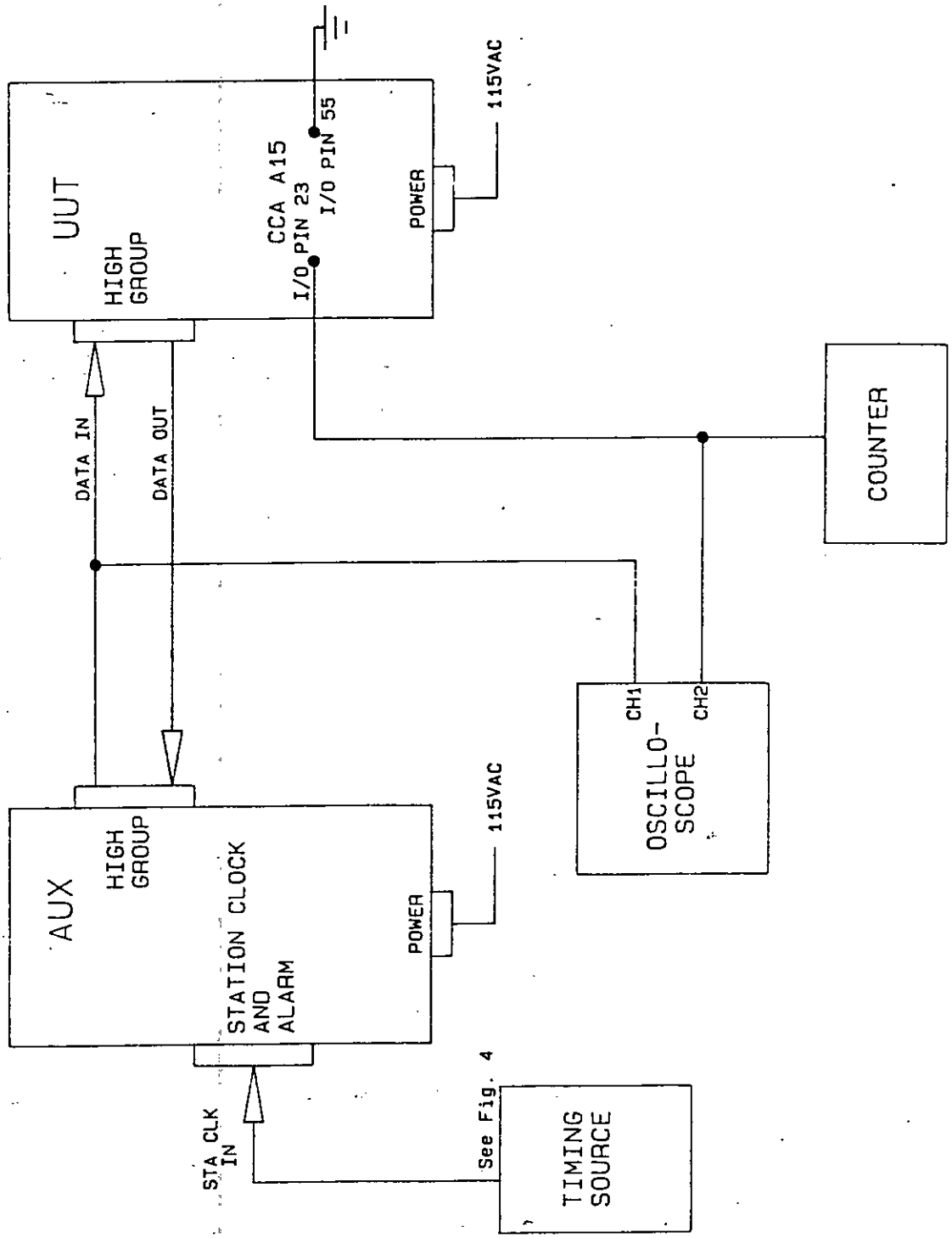


FIGURE 11. Timing modes.

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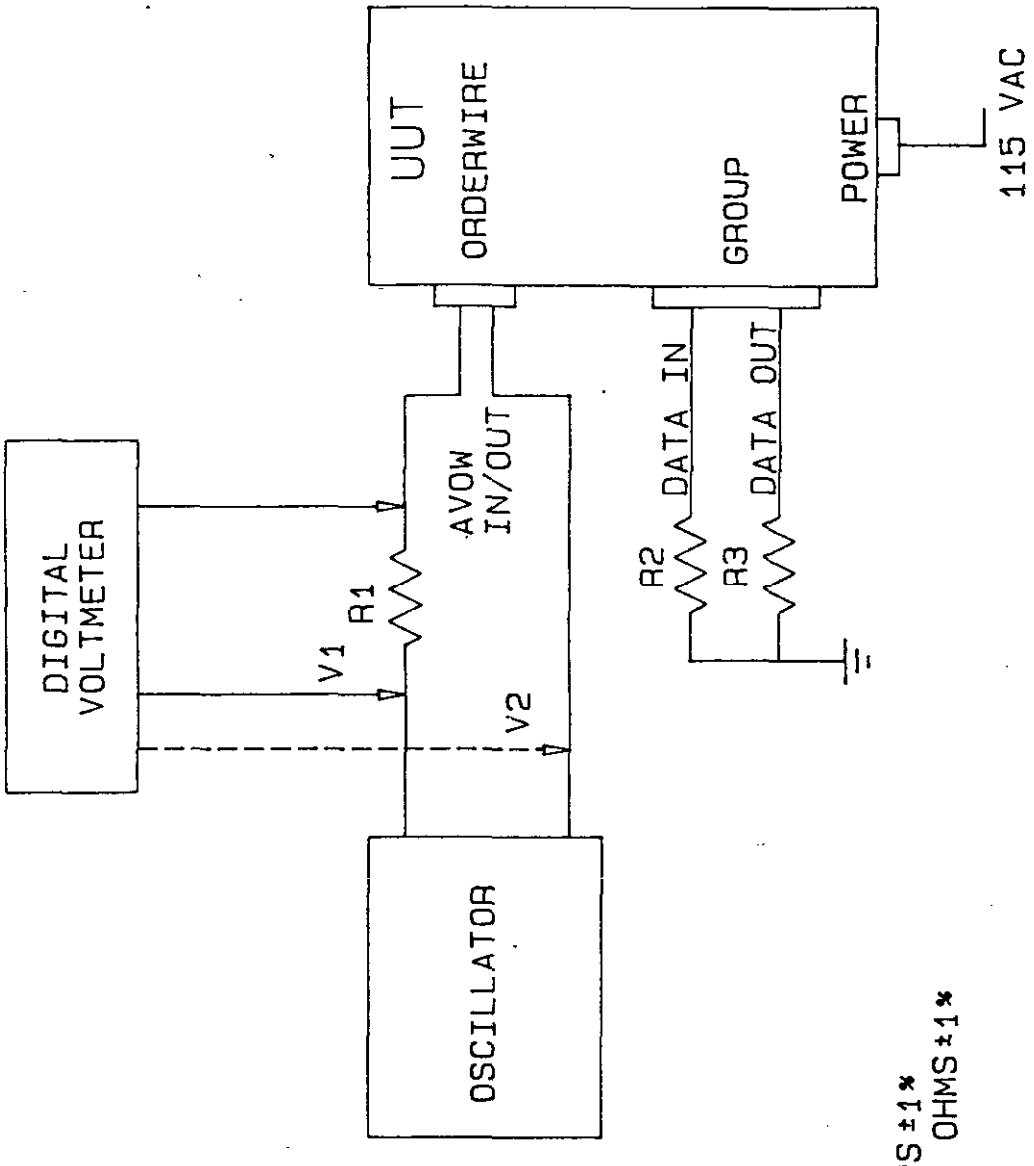
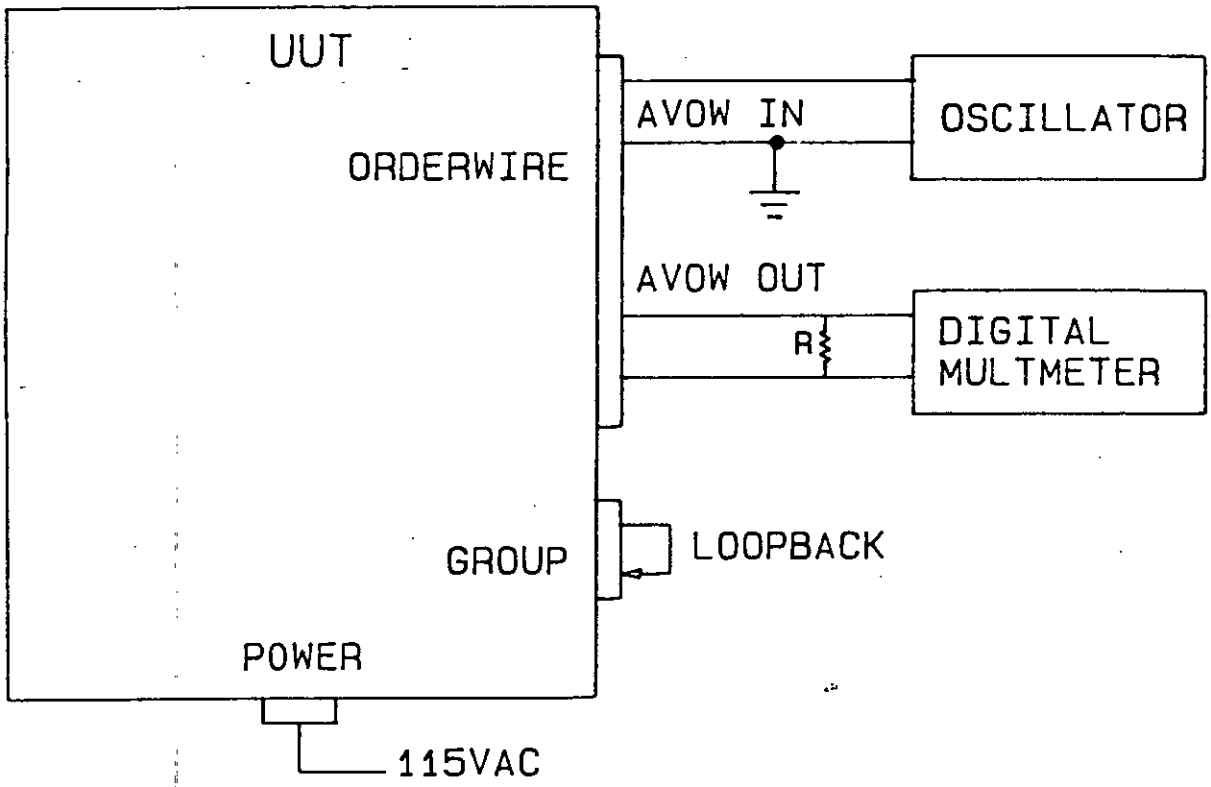


FIGURE 12. Orderwire input/output impedance.

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R 600 OHMS ±1%

FIGURE 13. Bandwidth.

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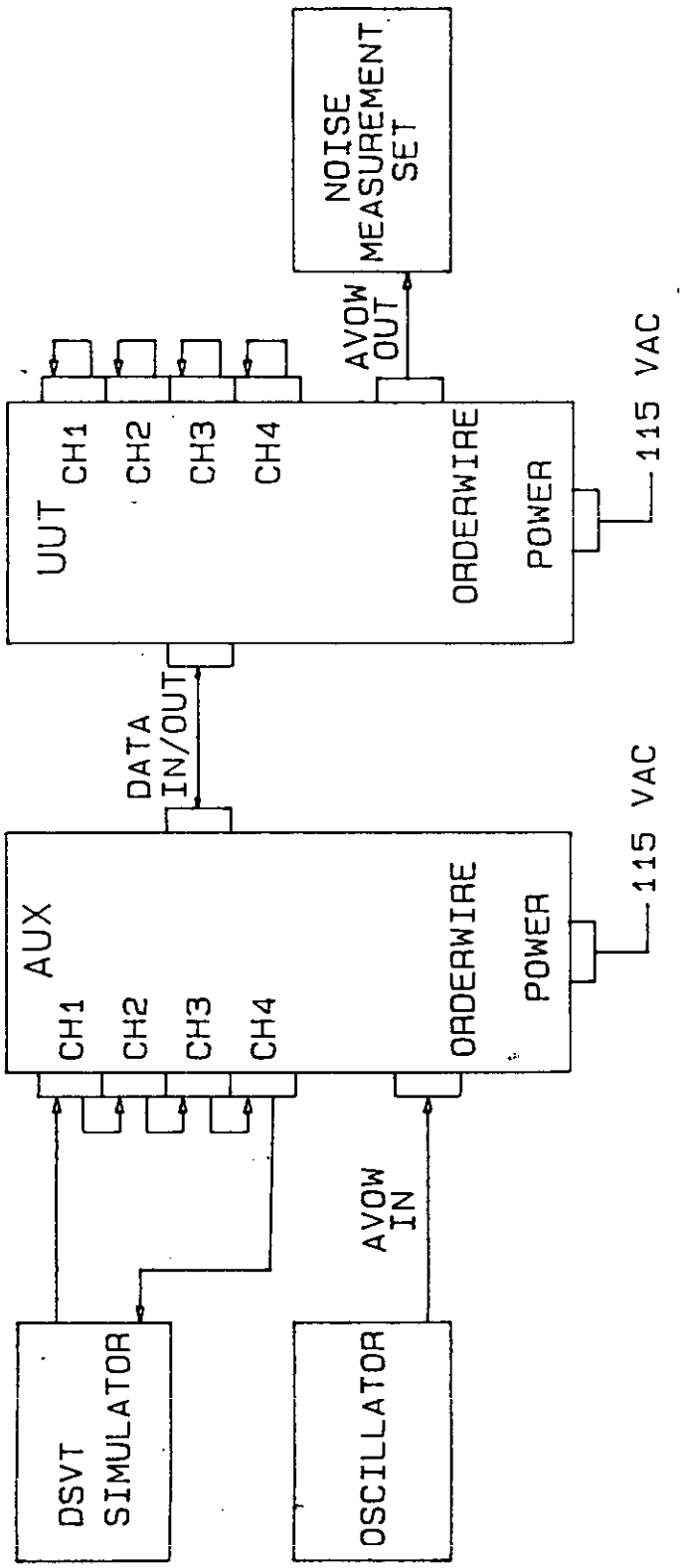
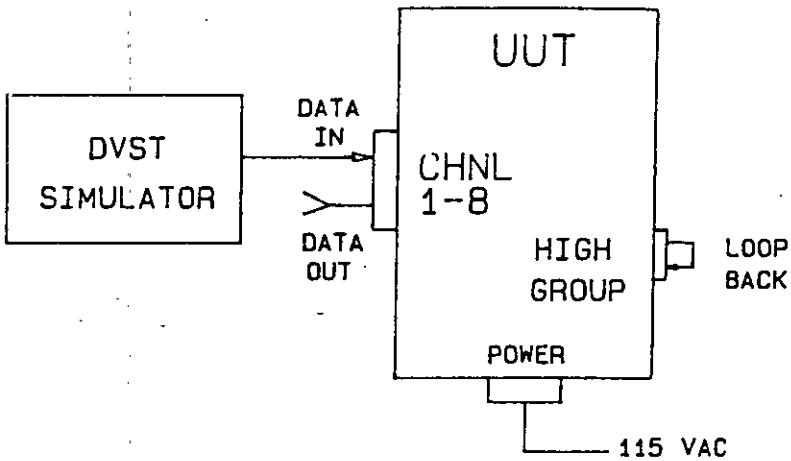
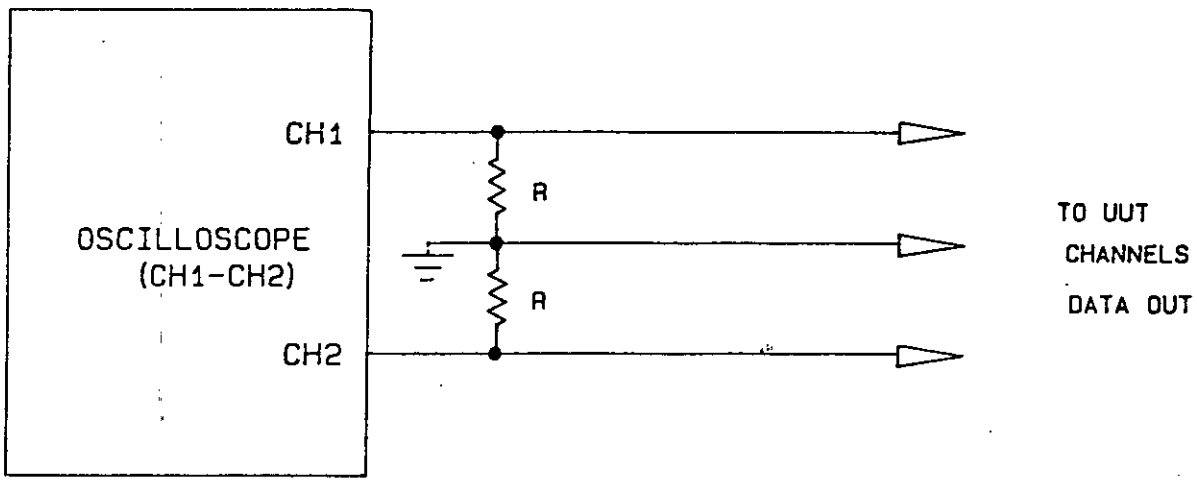


FIGURE 14. Signal-to-noise ratio.

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R 62.5 OHMS ±1%

FIGURE 15. Output voltage/wave shape.

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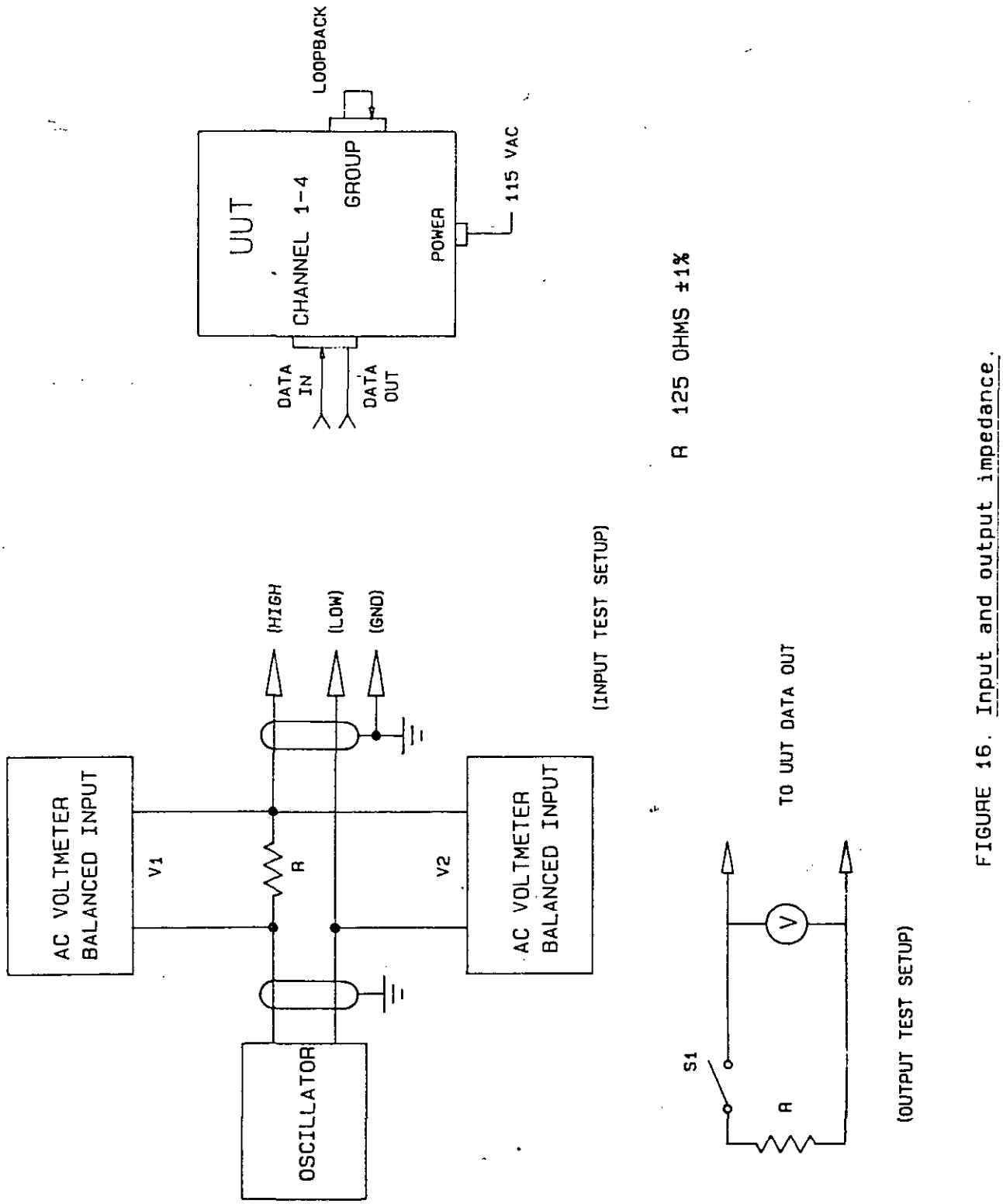


FIGURE 16. Input and output impedance.

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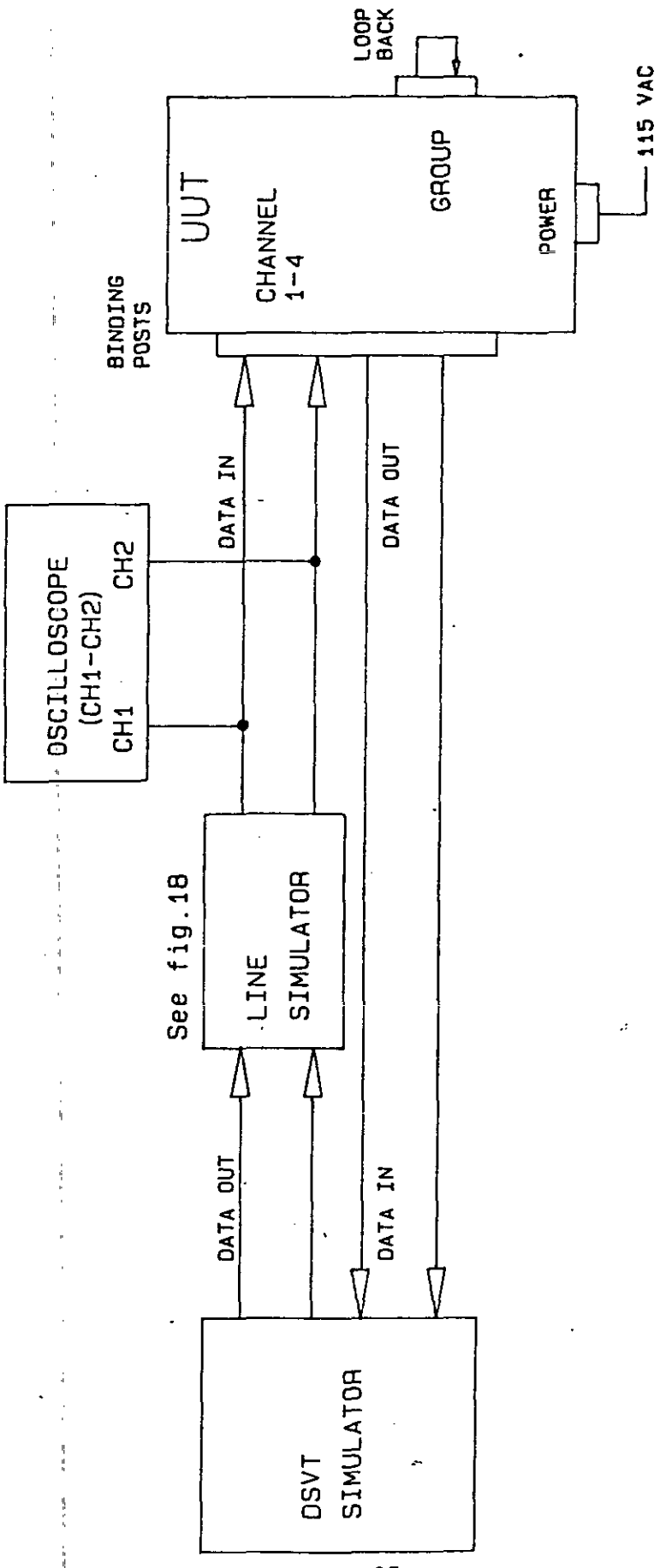


FIGURE 17. Input voltage.

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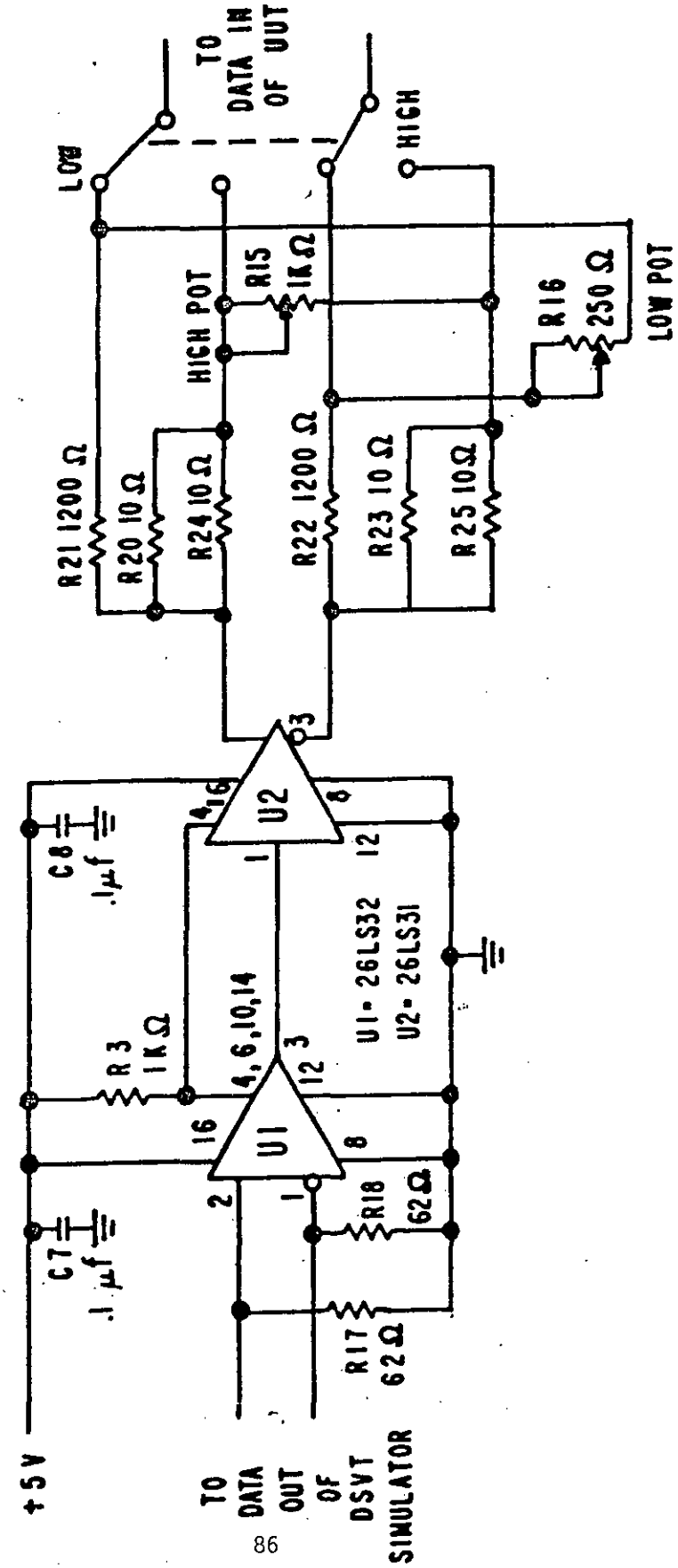


FIGURE 18. Line Simulator detail.

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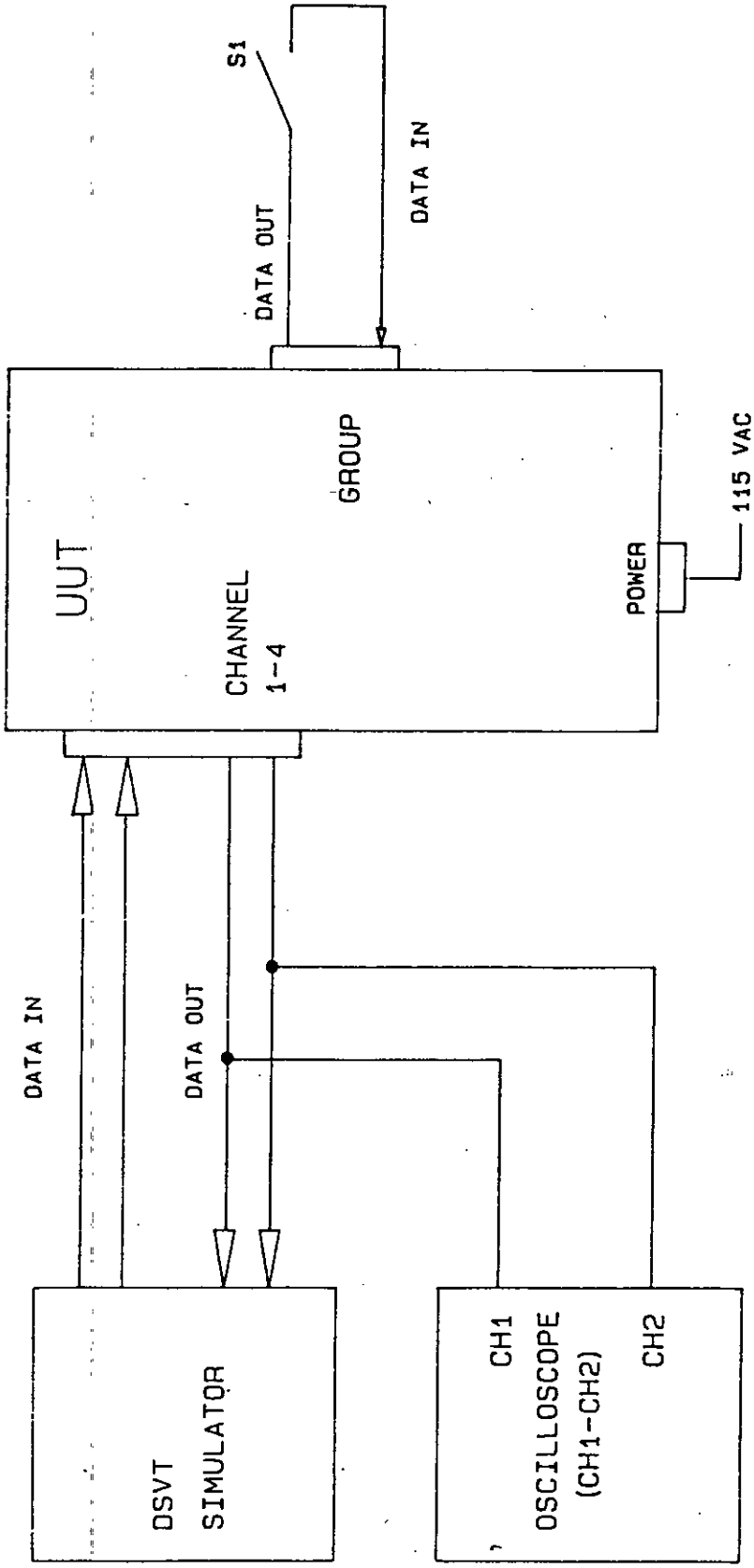
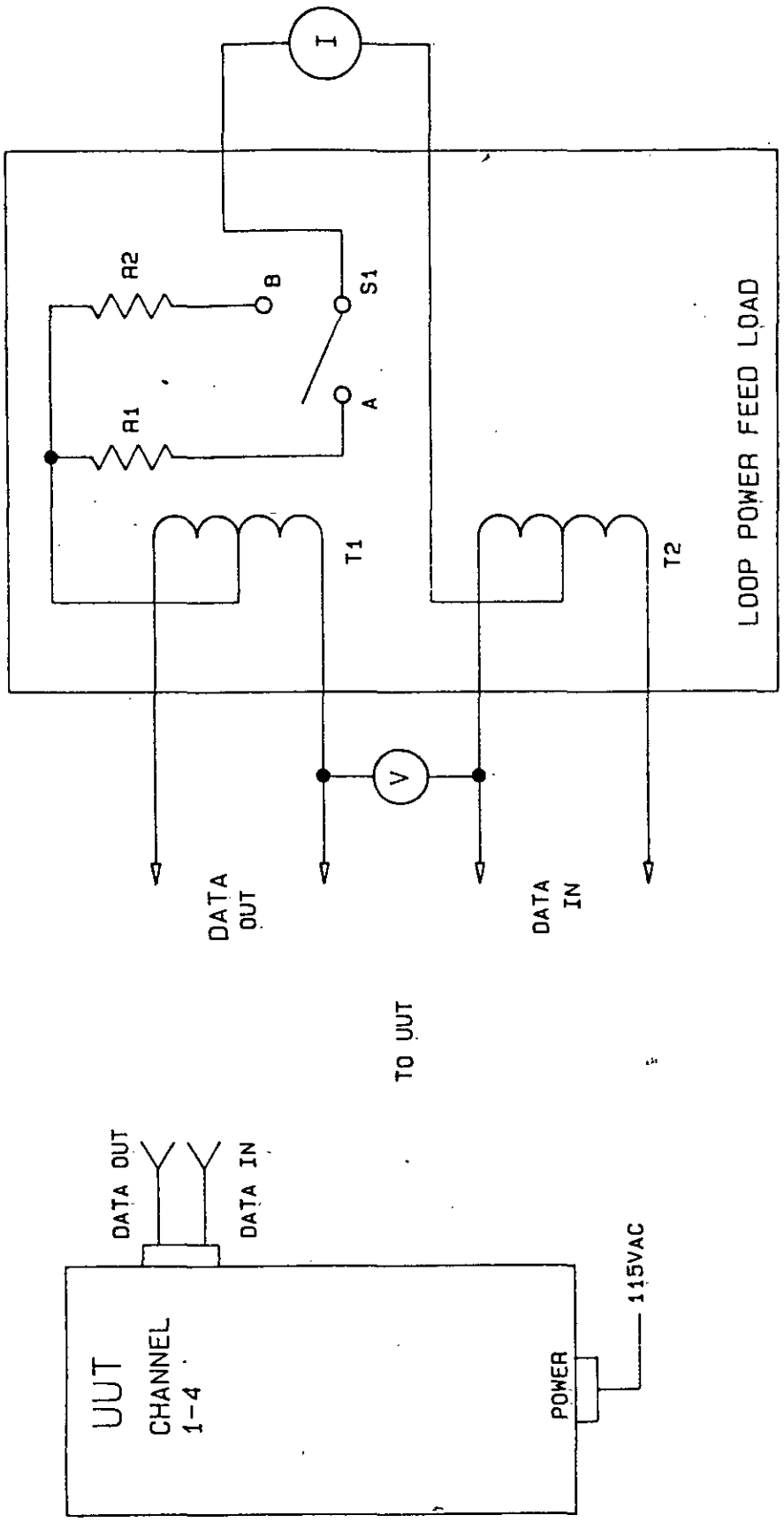


FIGURE 19. Idle loop; idle filler.

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NOTE: FOR T1 AND T2 SEE SM-A-985920
R1 11.2K ±1% 0.5W
R2 680 OHMS ±1% 5W

FIGURE 20. Loop power feed load.

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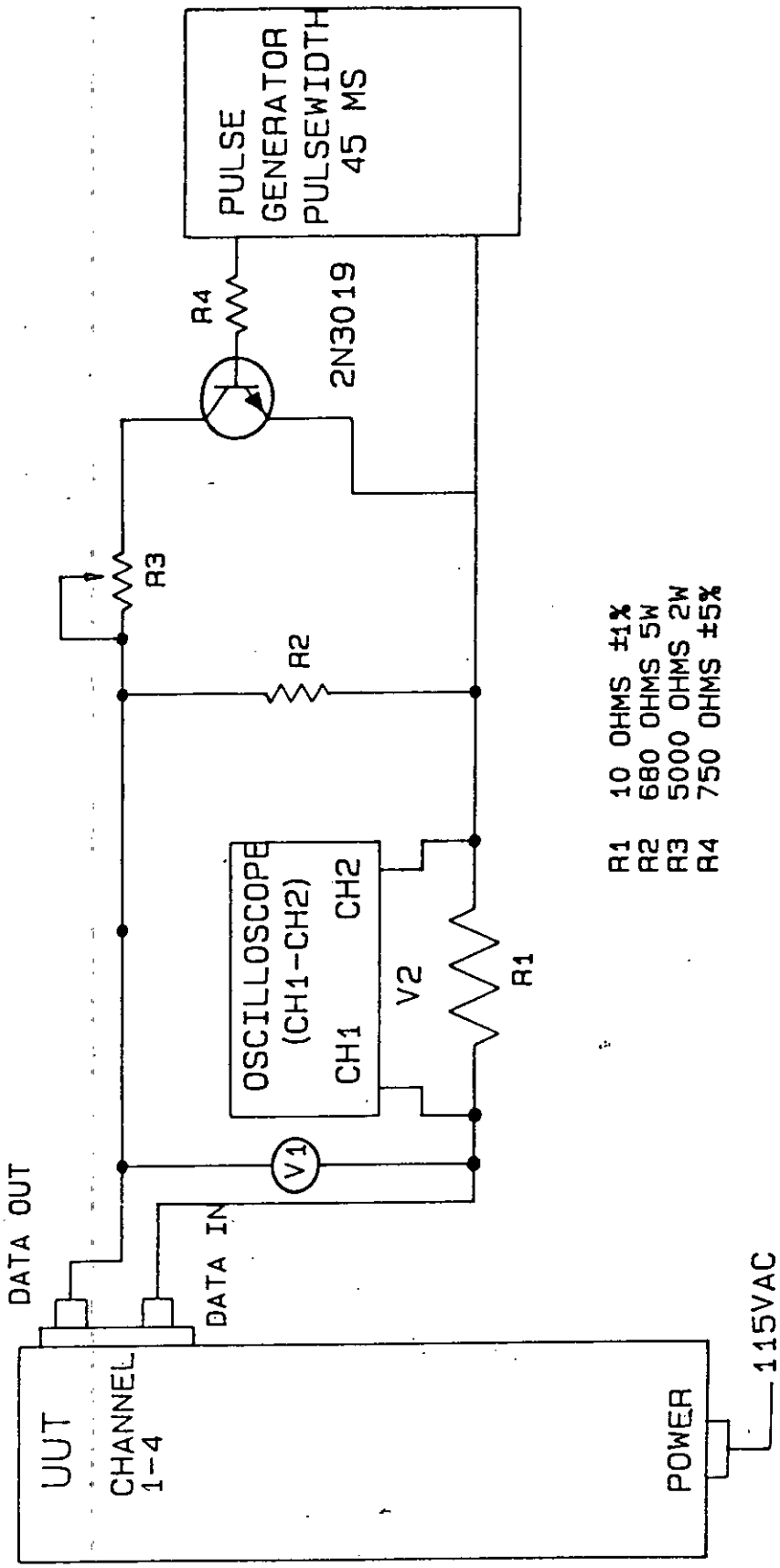


FIGURE 21. Loop power surge.

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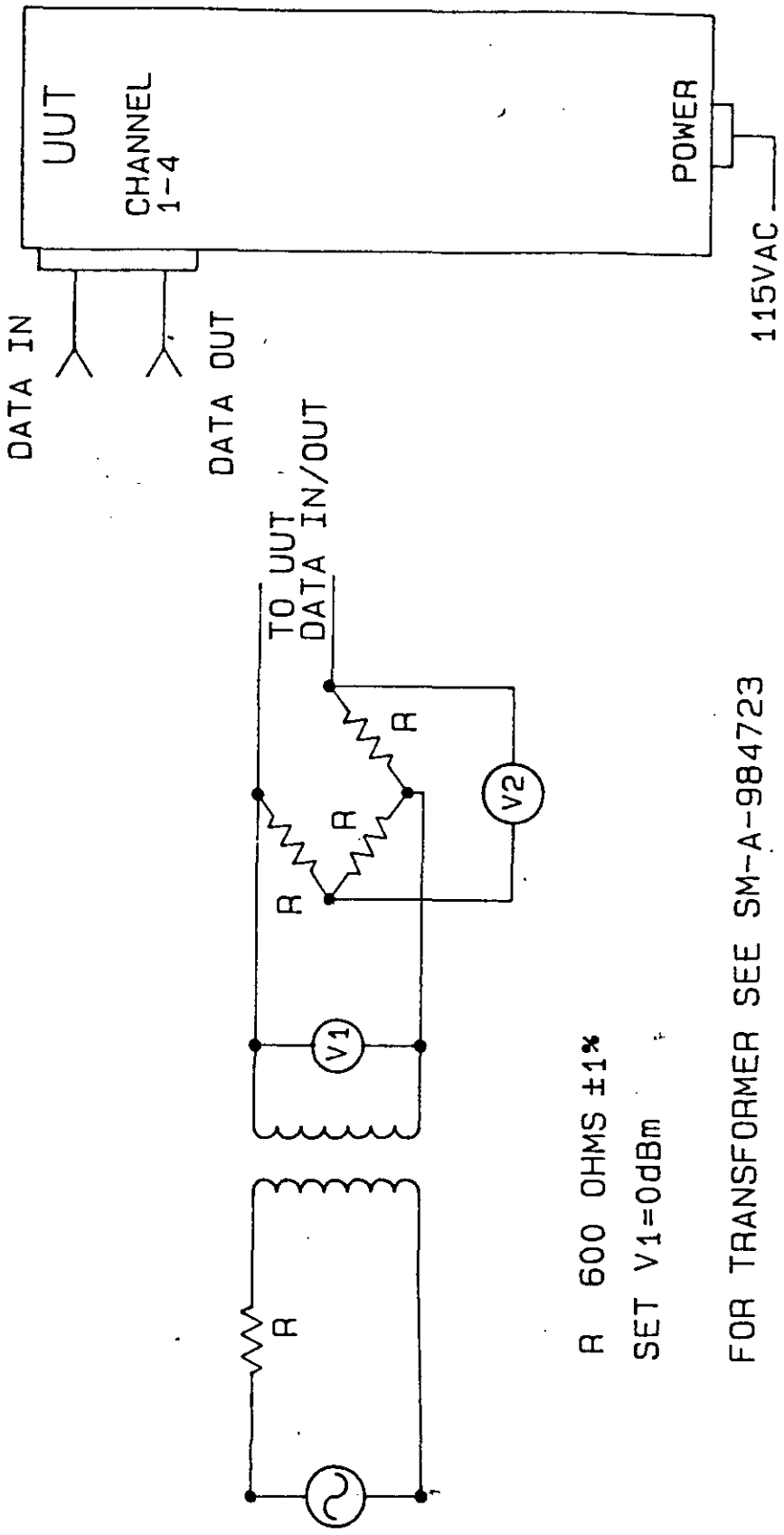
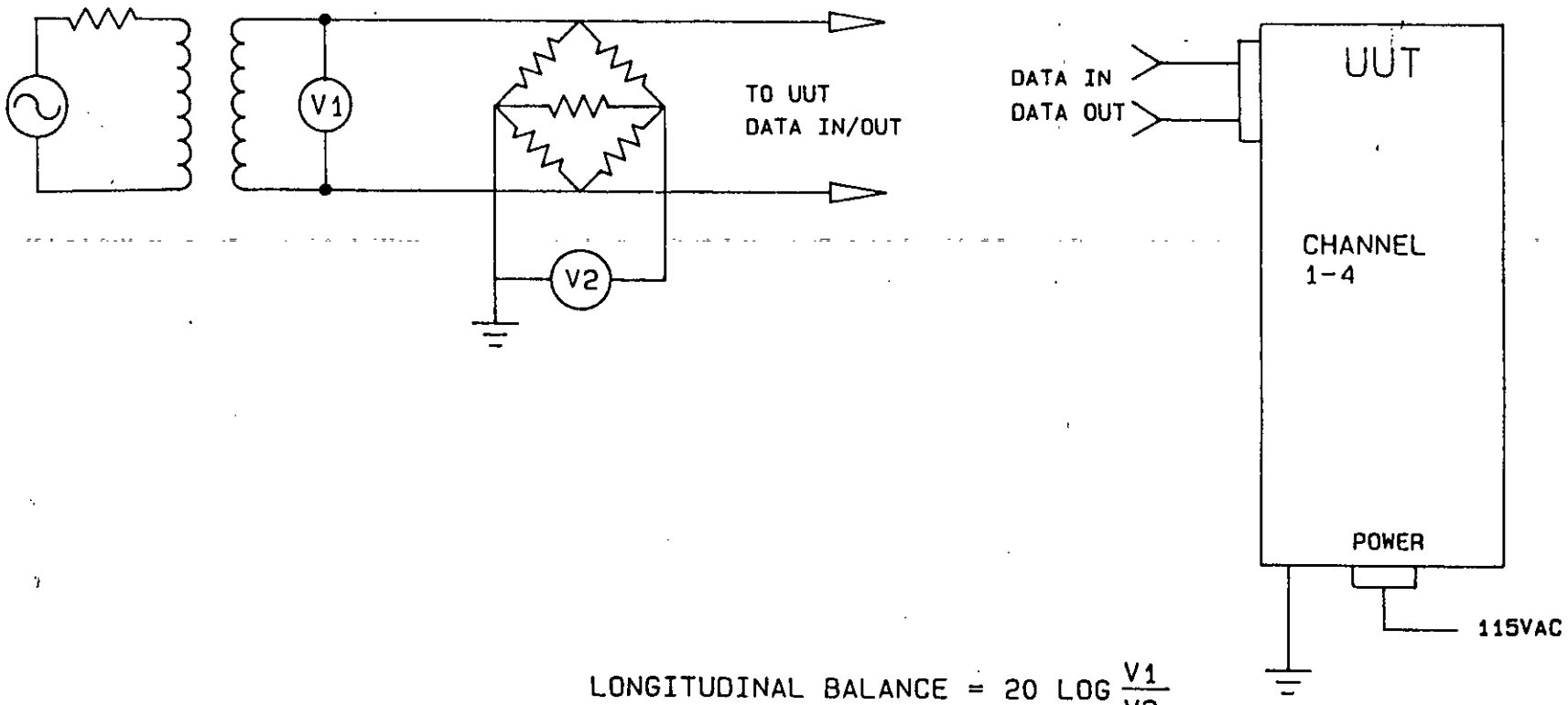


FIGURE 22. Impedance (return loss)

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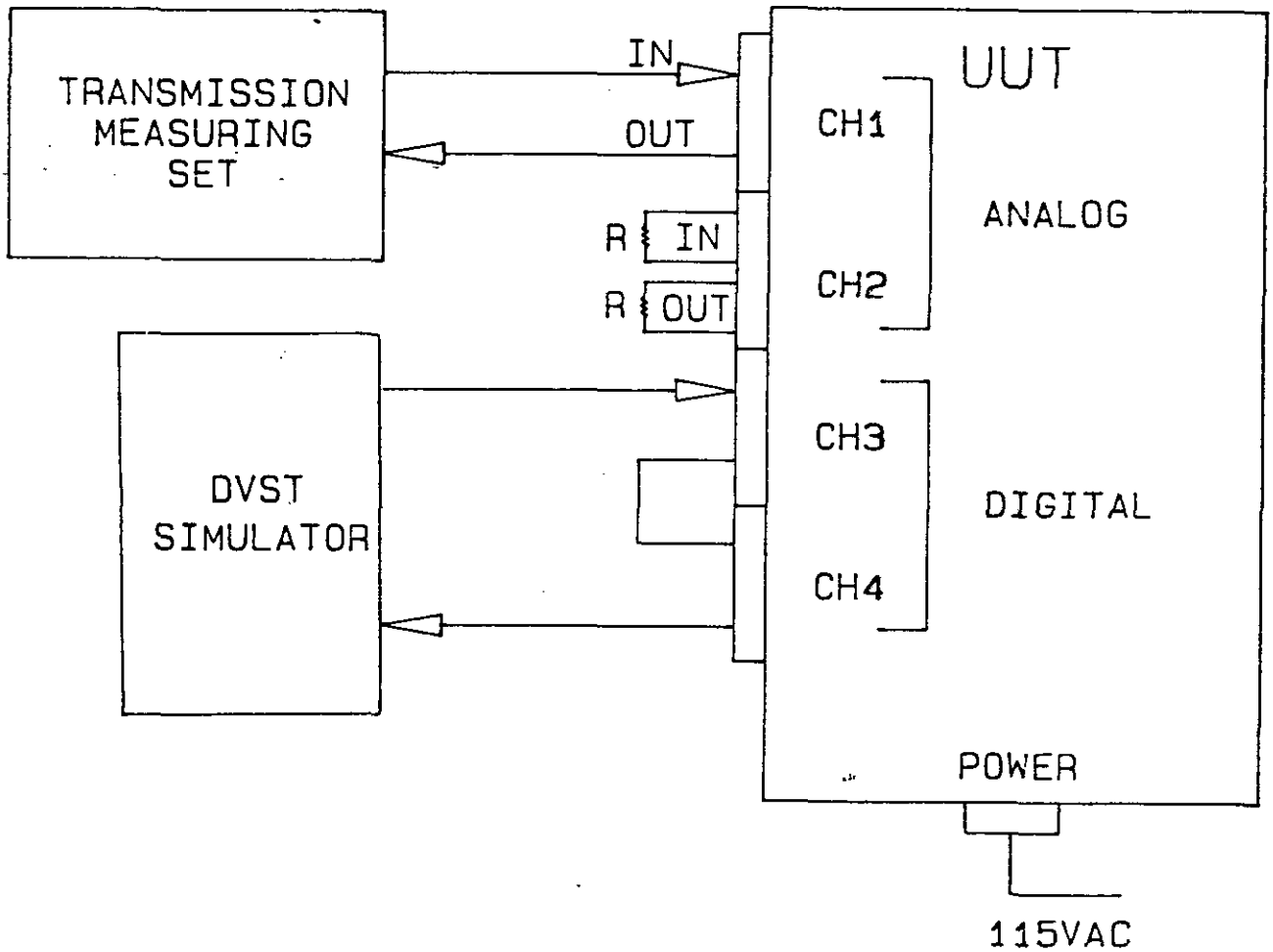


$$\text{LONGITUDINAL BALANCE} = 20 \log \frac{V1}{V2}$$

ALL RESISTORS 600 OHMS 1%
BRIDGE MUST BE TRIMMED SO THAT
BALANCE IS GREATER THAN 60 db
WITH NO OUTPUT CONNECTED.
UUT CASE GROUND MUST BE CONNECTED TO TEST CIRCUIT GROUND.
FOR TRANSFORMER SEE SM-A-984723

FIGURE 23. Longitudial balance.

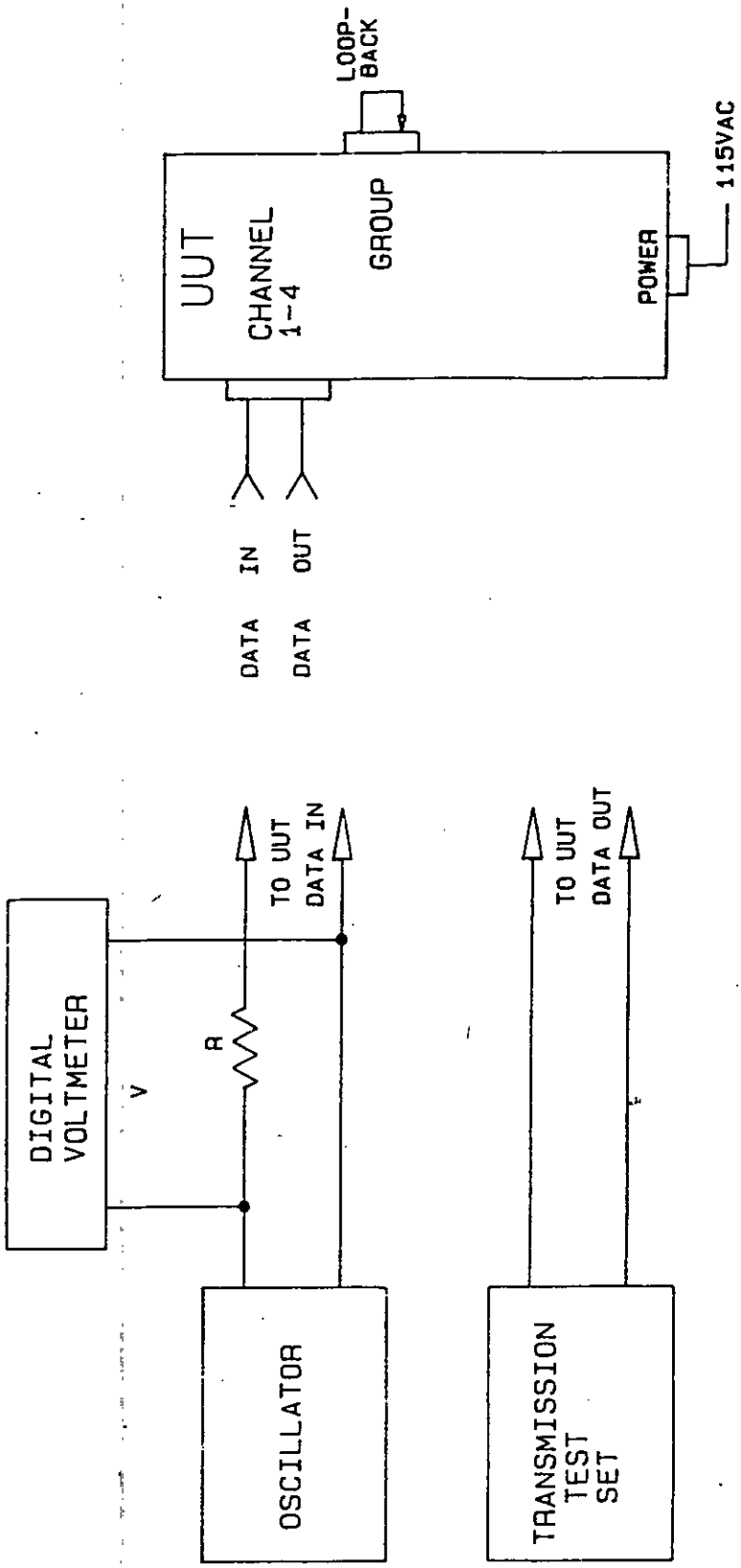
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R 600 OHMS ± 1%

FIGURE 24. Idle channel noise.

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R 600 OHMS ±1%, 0.5 W

FIGURE 25. Crosstalk.

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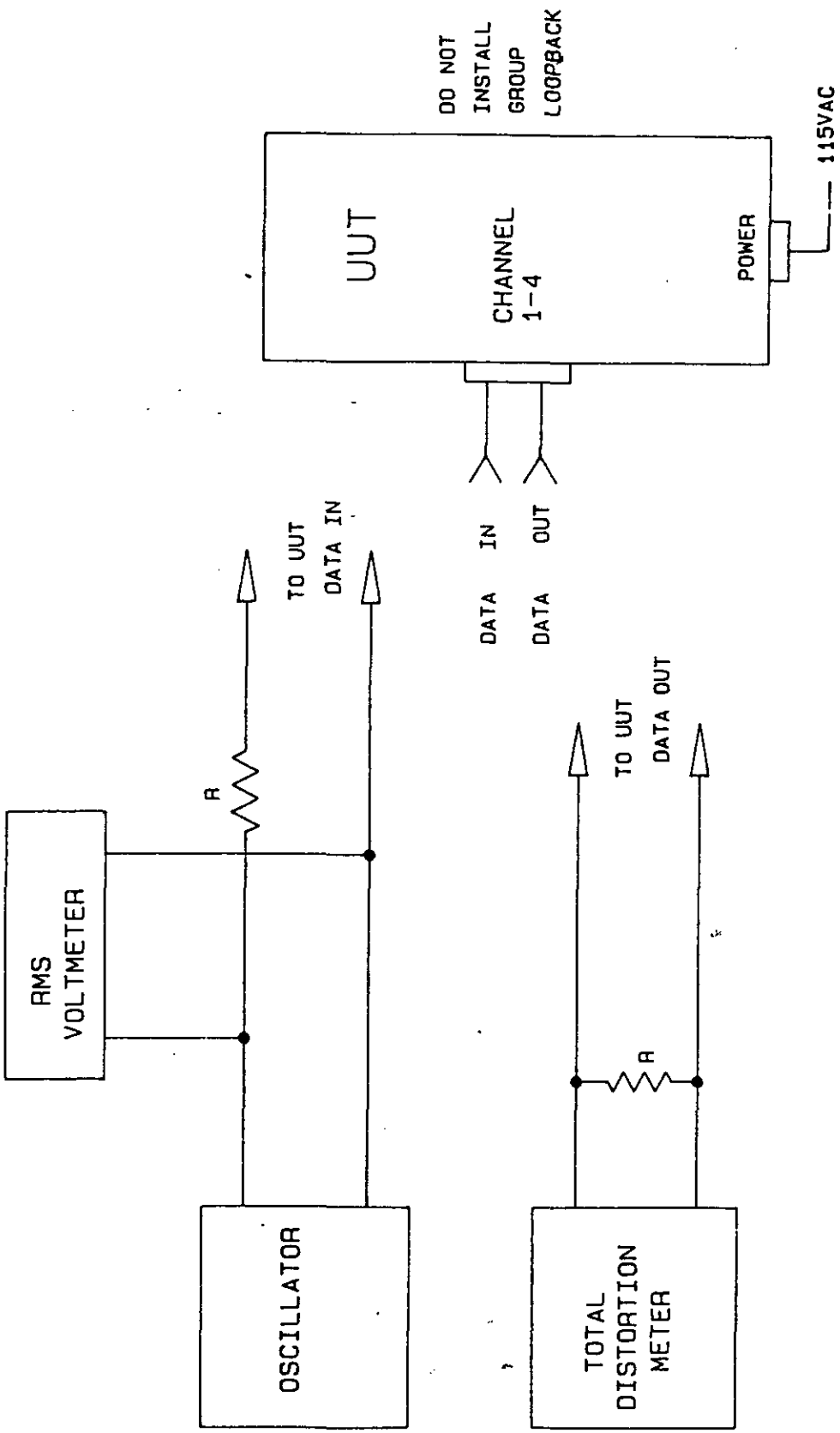


FIGURE 26. Peak clipping.

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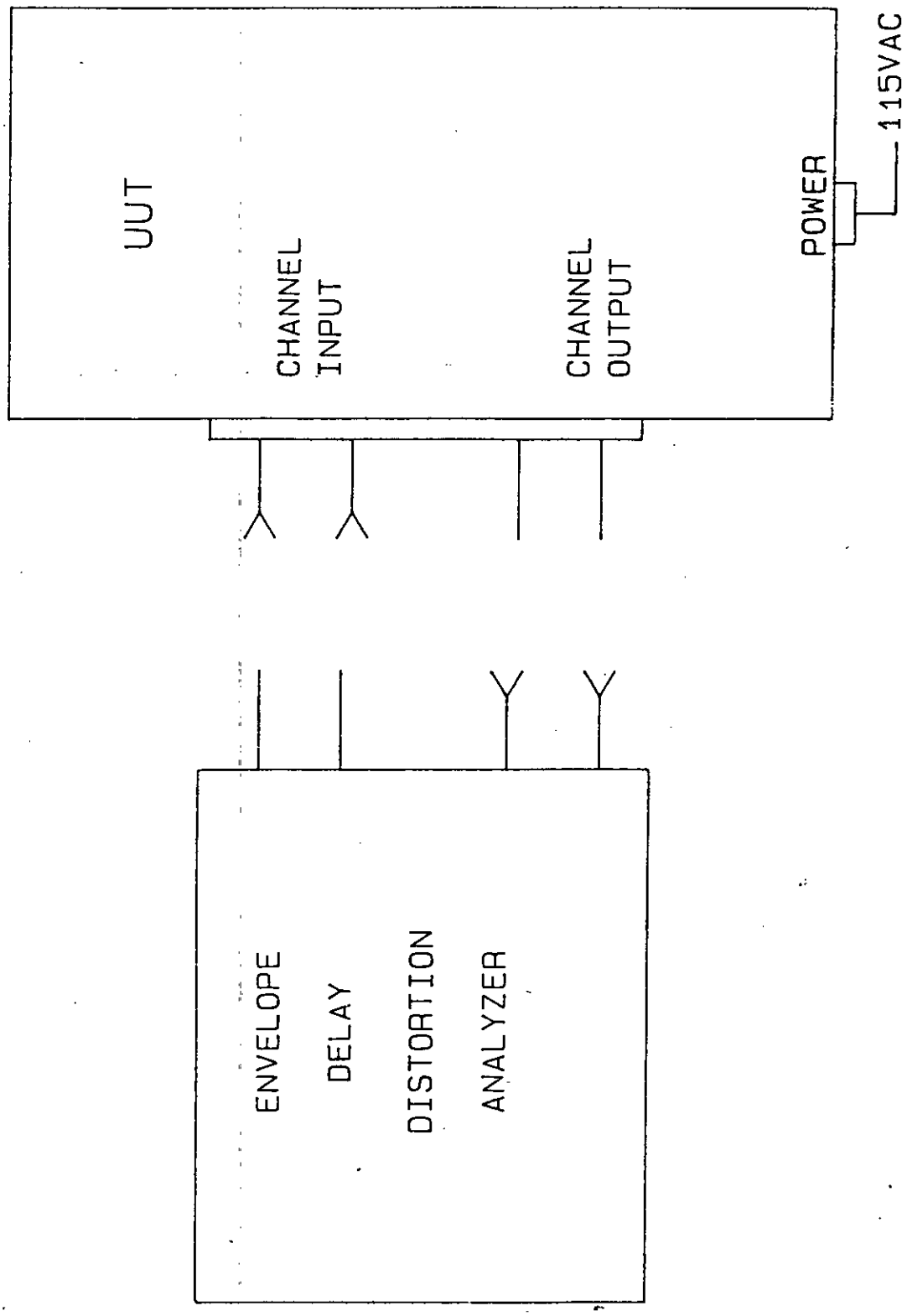


FIGURE 27. Envelope delay distortion.

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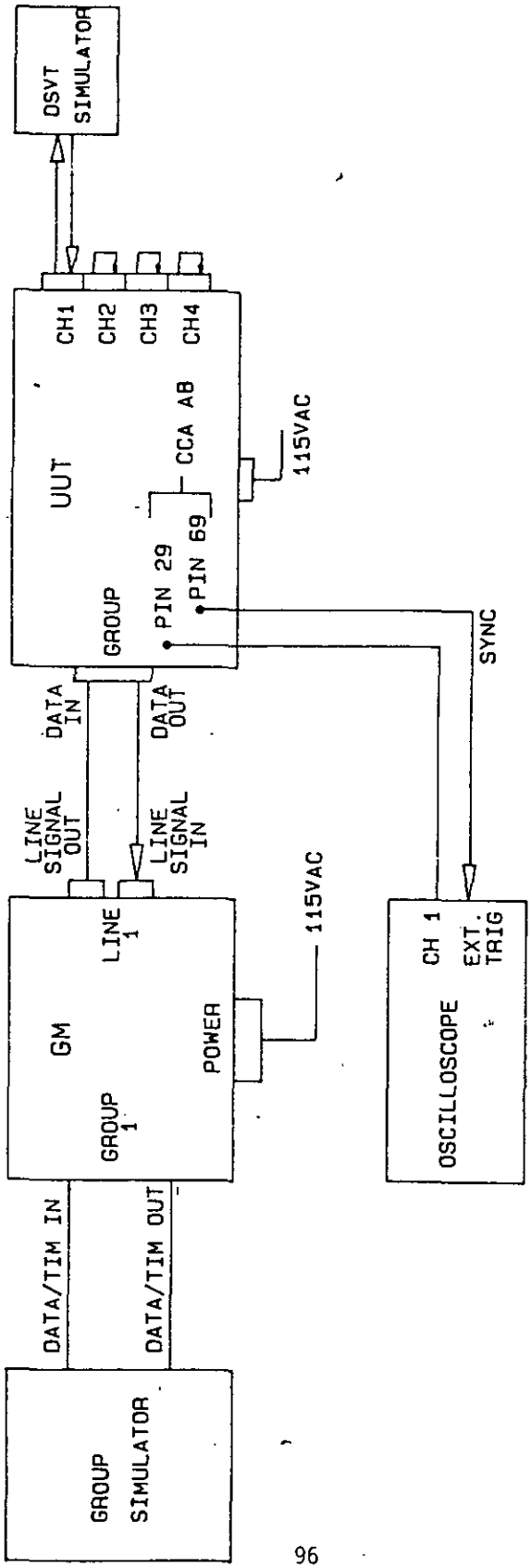


FIGURE 28. Format.

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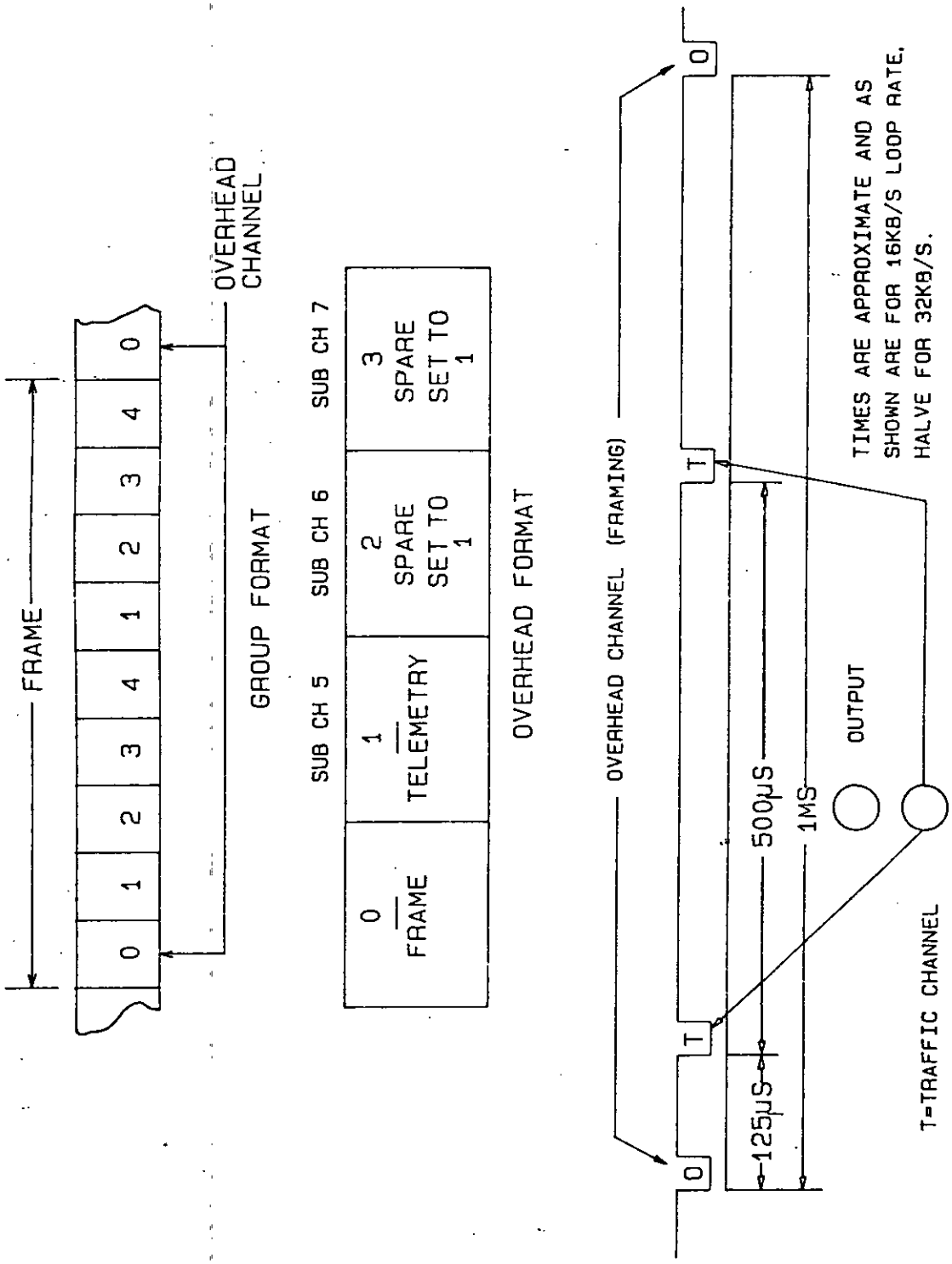


FIGURE 29. TD-1233 formats.

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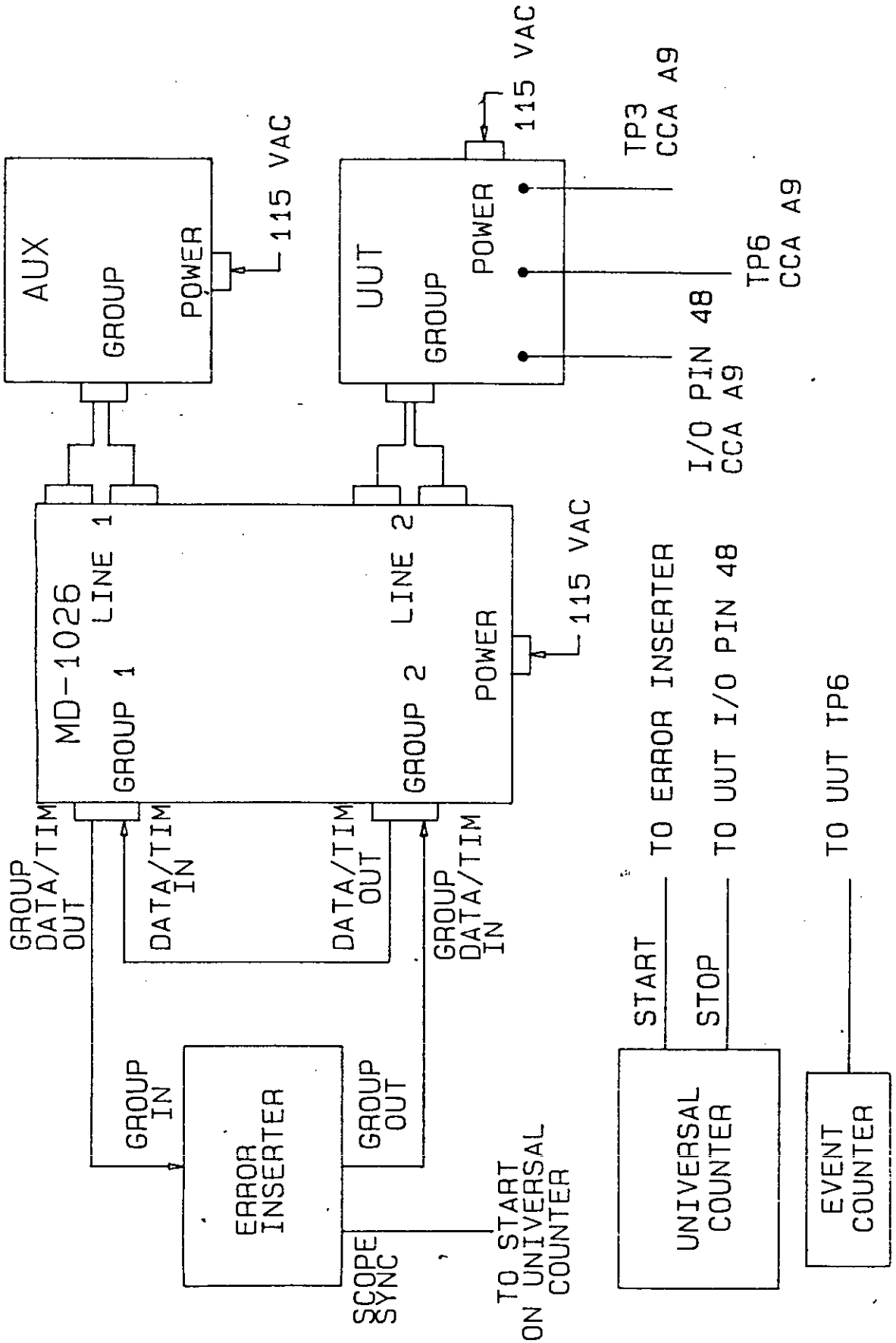
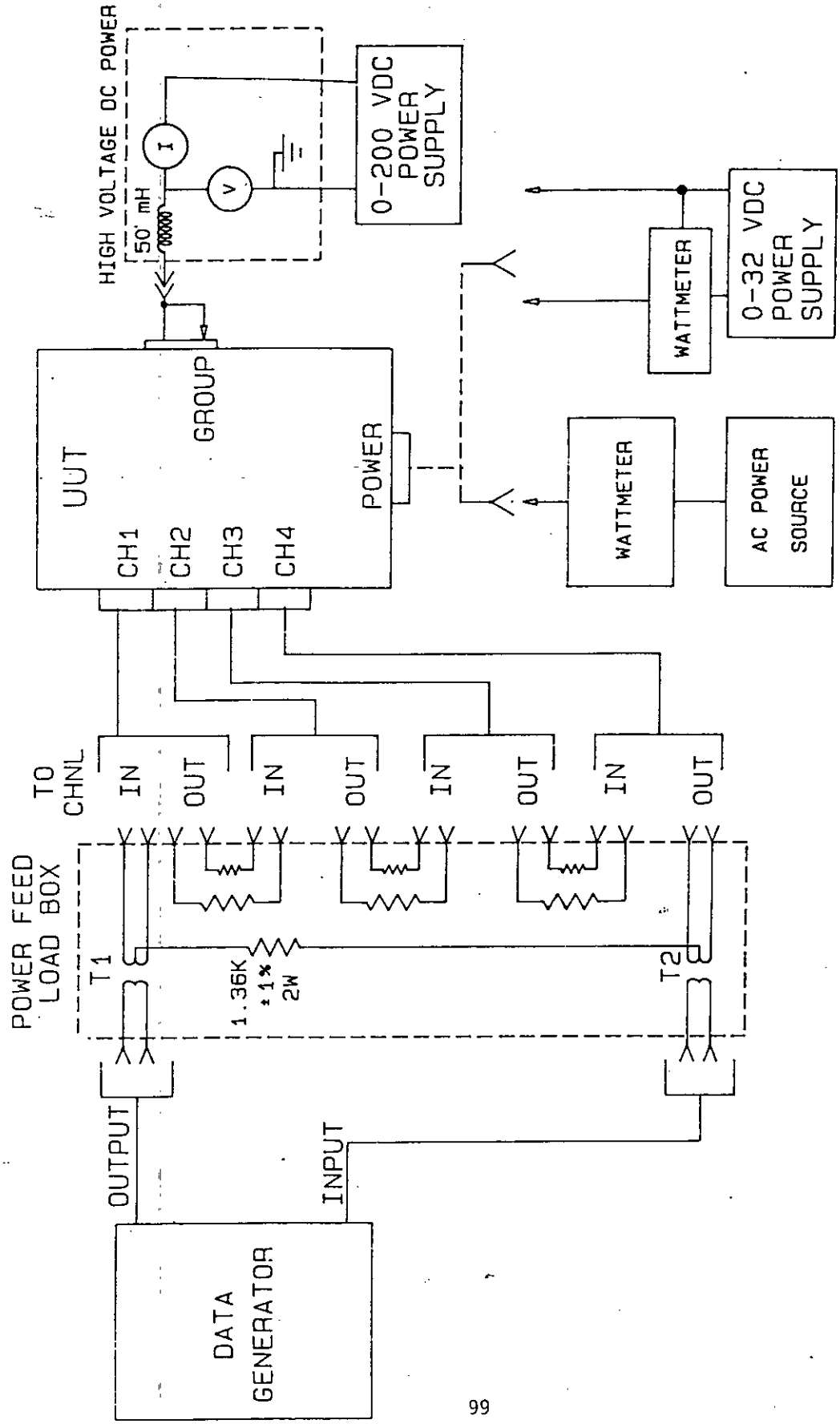


FIGURE 30. Frame acquisition; maintenance; detection of out-of-sync.

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ALL RESISTORS 2.72K ±1%, 2W, EXCEPT AS SHOWN.
T1 AND T2 SEE SM-A-985920.

FIGURE 31. Prime power.

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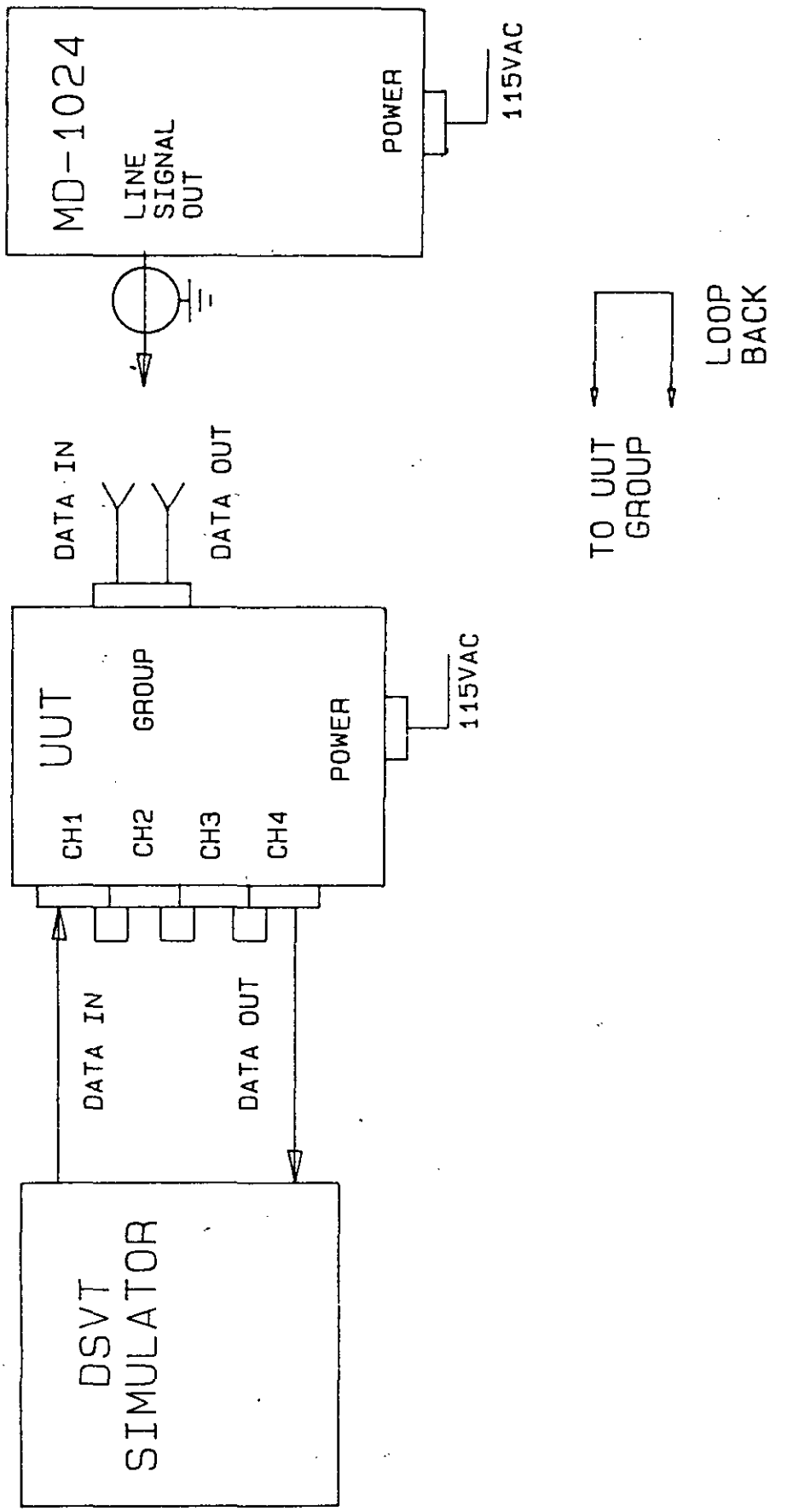


FIGURE 32. Current protection.

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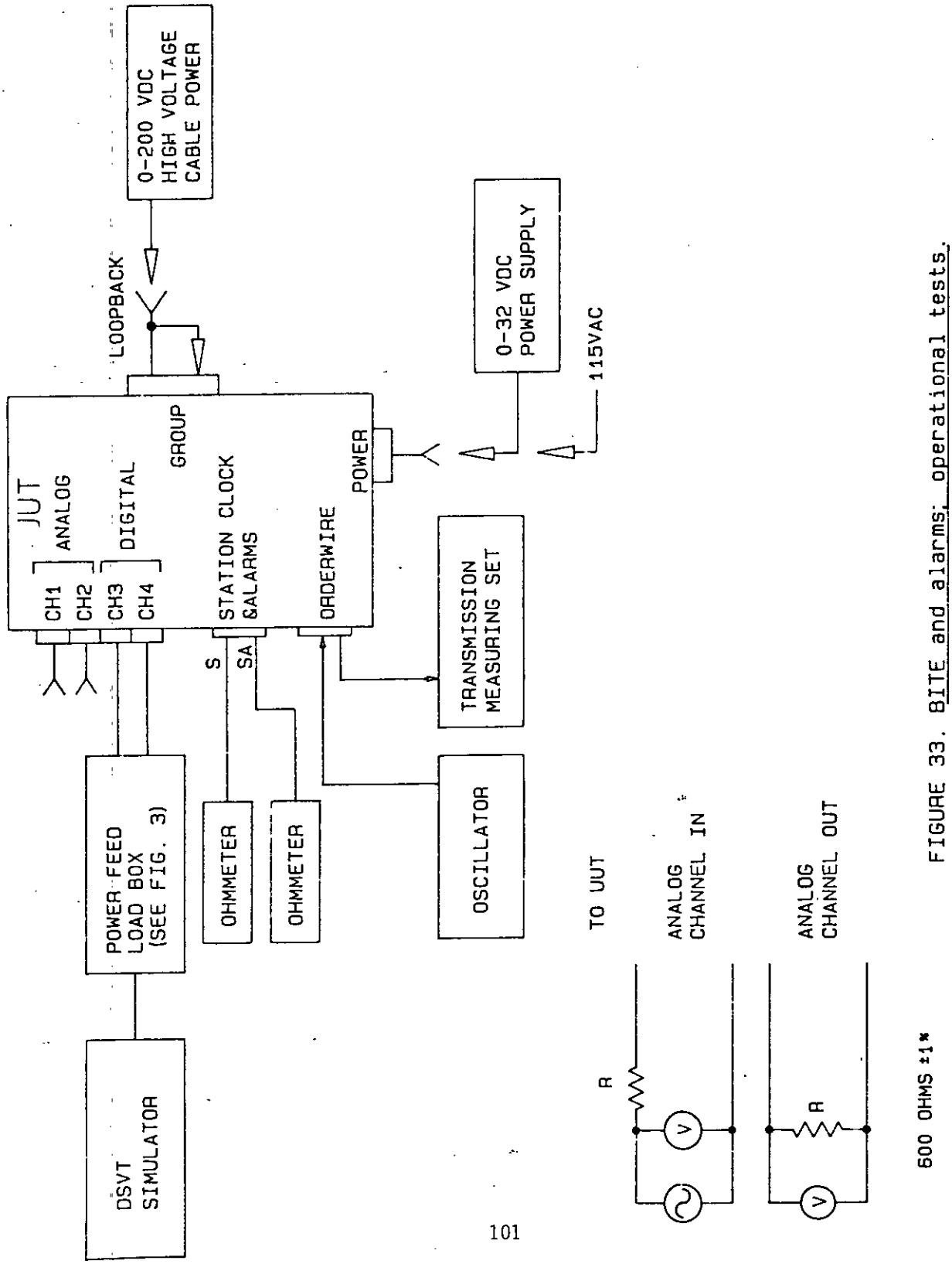


FIGURE 33. BITE and alarms; operational tests.

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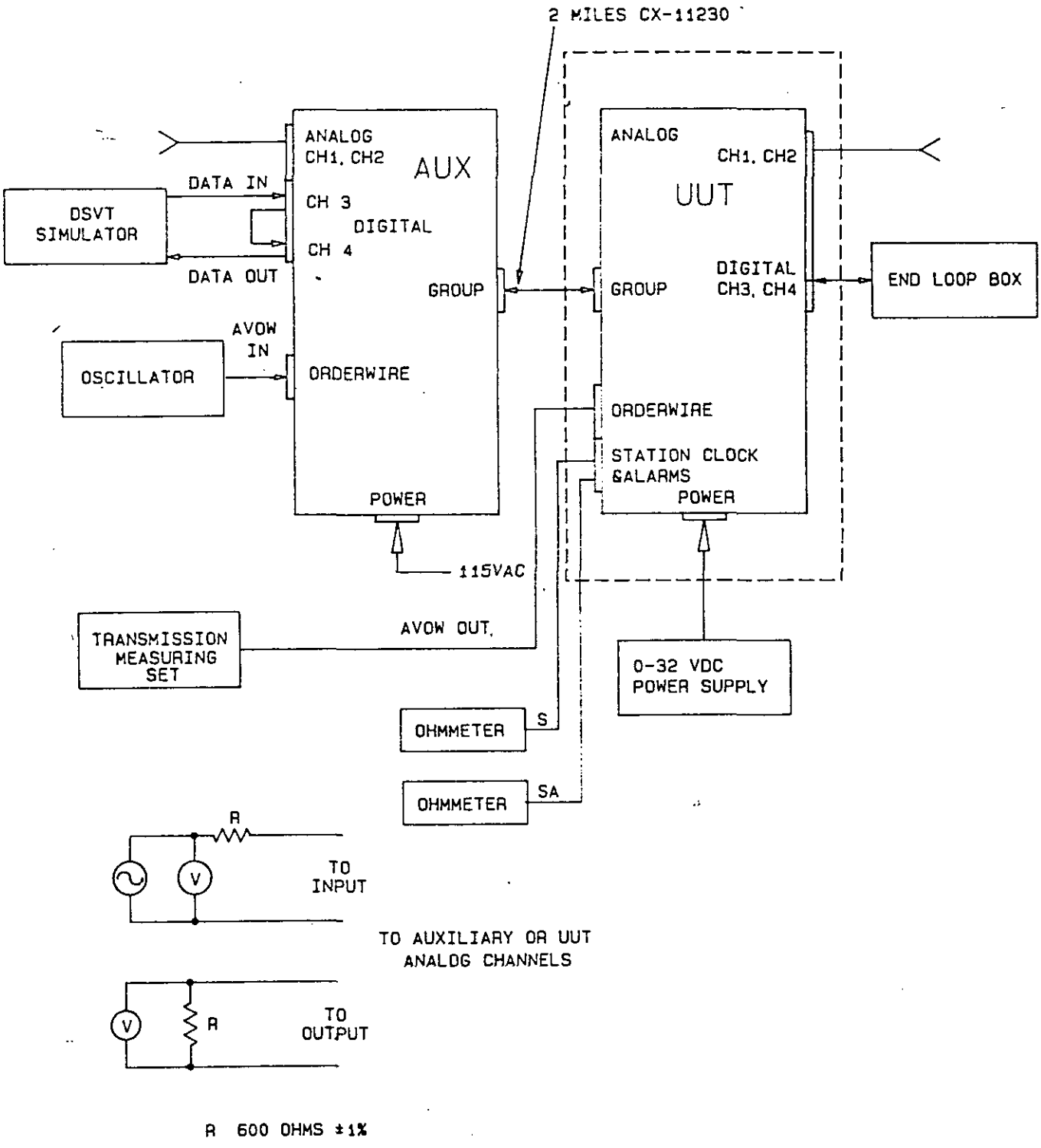


FIGURE 34. Environmental operational test configuration #1.

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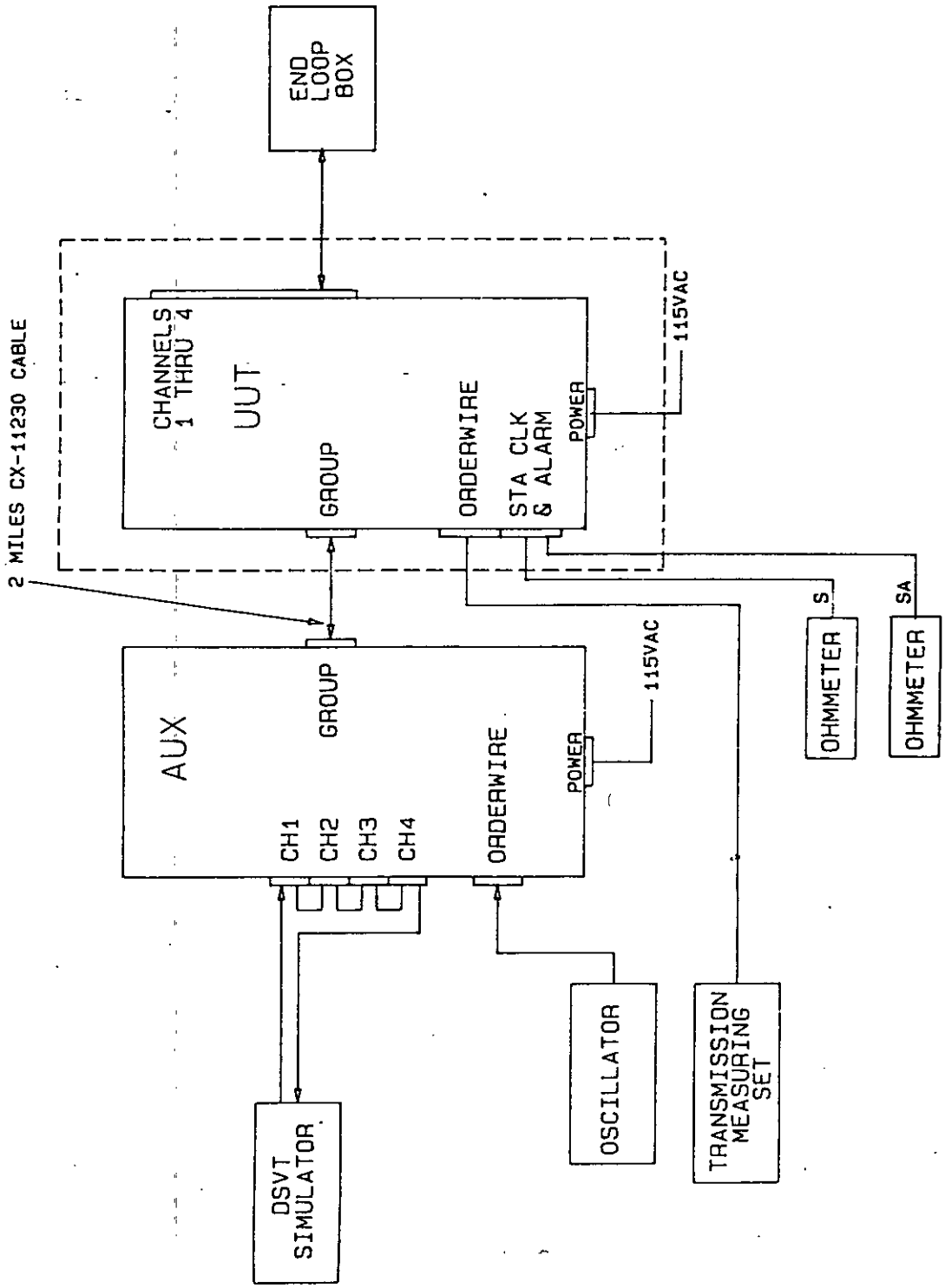


FIGURE 35. Environmental operational test configuration #2.

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